*BY ORDER OF THE COMMANDER*
*AIR FORCE GLOBAL STRIKE COMMAND*




*AIR FORCE GLOBAL STRIKE COMMAND INSTRUCTION 21-105*

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*Maintenance*

# *CORROSION PREVENTION AND CONTROL PROGRAM*

**COMPLIANCE WITH THIS PUBLICATION IS MANDATORY**

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This instruction implements Air Force Policy Directive (AFPD) 21-1, *Maintenance of Military Material*, and is consistent with AFPD 13-5, *Air Force Nuclear Enterprise*. It references Department of the Air Force Instruction (DAFI) 21-101, *Aircraft and Equipment Maintenance Management*, and Air Force Manual (AFMAN) 21-202, *Missile Maintenance Management*, and establishes policy and assigns responsibility to establish and support the Corrosion Prevention and Control Program for Air Force Global Strike Command (AFGSC). This instruction applies to AFGSC Regular Air Force, Air Force Reserve Command (AFRC) and to the Air National Guard (ANG) Classic Associate units associated with AFGSC. It does not apply to United States Space Force. This publication may be supplemented at any level, but all supplements must be routed to the Office of Primary Responsibility (OPR) of this publication for coordination prior to certification and approval. The authorities to waive wing, and unit level requirements in this

publication are identified with a tier number ("T-0, T-1, T-2, T-3") following the compliance statement. See Department of the Air Force Manual (DAFMAN) 90-161, *Publishing Processes and Procedures*, for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or alternately, to the Publication OPR for non-tiered compliance items. Ensure that all records created as a result of processes prescribed in this publication are maintained In Accordance With (IAW) AFI 33-322, *Records Management and Information Governance Program*, and disposed of IAW the Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS). Refer recommended changes and questions about this publication to the OPR, AFGSC/A4MQ, using the Department of the Air Force (DAF) Form 847, *Recommendation for Change of Publication*; route DAF Form 847s from the field through the appropriate functional chain of command to **AFGSC.A4M.workflow@us.af.mil**.

**(DYESSAFB)** AFGSCI21-105, *Corrosion Prevention and Control Program,* is supplemented as follows: This instruction establishes policy and assigns responsibility to establish and support the Corrosion Prevention and Control Program for Air Force Global Strike Command (AFGSC) in accordance with (IAW) AFGSCI 21-105, *Corrosion Prevention and Control Program,* dated 17 June 2022. This instruction does not apply to Air Force Reserve Command (AFRC) or to the Air National Guard (ANG); however, procedures in this instruction will be followed by AFRC/ANG Classic Associate units associated with AFGSC. Ensure that all records created as a result of processes prescribed in this publication are maintained IAW AFI 33-322, *Records Management and Information Governance Program* and are disposed of IAW the Air Force Records Disposition Schedule (RDS) located in the Air Force Records Information Management System (AFRIMS). Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the DAF Form 847, *Recommendation for Change of Publication*; route DAF Form 847s from the field through the appropriate functional chain of command.

## *SUMMARY OF CHANGES*

This publication has been substantially revised and must be completely reviewed in its entirety. Major changes include mural guidance for Missile Alert Facilities, Nose Art routing procedures, adds MH-139 helicopter guidance, updates specific markings and paint colors for aircraft and missiles, changes specific responsibilities for Corrosion Control and Prevention Managers and the addition of facility requirements in accordance with AFI 32-1024, *Standard Facility Requirements*.

**(DYESSAFB)** Minor changes in this publication include, but are not limited to, referral to AFGSCI 21-105 regarding tone-down procedures and the use of Defense Property Accountability System (DPAS) to schedule and document AGE painting. This document should be reviewed thoroughly.

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## Chapter 1

## ROLES AND RESPONSIBILITIES

**1.1. Headquarters (HQ) AFGSC/A4M Responsibilities.** HQ AFGSC/A4M will designate the AFGSC Fabrication Functional Manager or Low Observable (LO) Functional Manager as the Command Aircraft Corrosion Prevention and Control Program Manager. The program manager will:

1.1.1. Manage the Corrosion Prevention and Control Program for all assigned aircraft.

1.1.2. Coordinate all nose art request packages thru AFGSC/A1 and JA prior to submission to AFGSC/A4 for final approval. Once approved, package will be forwarded to AFGSC/HO and applicable unit historian for archive.

1.1.3. Represent command at assigned weapon systems Corrosion Prevention Advisory Boards (CPAB), AF/DoD corrosion conferences, field surveys, and other Environmental/Technical conferences as applicable.

1.1.4. Ensure adequate corrosion control training is available and current for all aircraft and Aerospace Ground Equipment (AGE) maintenance personnel.

1.1.5. Serve as the Command's interface with Air Education and Training Command (AETC) and the Air Force Institute of Technology (AFIT) for corrosion training course content changes.

1.1.6. Coordinate Major Command (MAJCOM) approval for test programs through appropriate Weapon System Team.

**1.2. Wing Commander Responsibilities.** The WG/CC will:

1.2.1. Approve all aircraft paint waivers and nose art requests. Ensure all requests have been routed through local Historian, Public Affairs, Wing Corrosion Manager, and Judge Advocate (JA) office. Submit all required documents to the HQ AFGSC/A4M, Aircraft Corrosion Prevention and Control Program Manager via Task Management Tool.

**1.3. Maintenance Group Commander Responsibilities.** The MXG/CC (or equivalent) will:

1.3.1. Ensure adequate corrosion control facilities are available to wash aircraft, perform minor maintenance, and paint assigned aircraft on a year-round basis. An outside wash rack may be used on an interim basis when weather conditions permit and when approved by Base Civil Engineer.

1.3.2. Ensure requirements outlined in UFC 4-211-02, *Corrosion Control and Paint Facilities*, and DAFMAN 32-1084, *Standard Facility Requirements,* are met for Support Equipment (SE) and aircraft small parts. **Note:** This capability can be incorporated in the aircraft corrosion control facility if space permits.

1.3.3. Ensure facility control technology meets local, state, and federal Environmental Protection Agency requirements in conjunction with current National Emission Standards for Hazardous Air Pollutants.

1.3.4. Ensure frequency of wash/rinse cycles are maintained in accordance with Technical Order (T.O.) 1-1-691, *Cleaning and Corrosion Prevention and Control, Aerospace and Non-Aerospace Equipment*, and revised as necessary based on changes in mission and location.

1.3.4.1. Ensure Plans, Scheduling & Documentation (PS&D) section schedule aircraft washes through applicable Maintenance Information System (MIS).

1.3.4.2. Ensure waiver requests are submitted for any assigned aircraft wash that will be overdue more than 30 days with an official memo to HQ AFGSC/A4M prior to the 30th day from the due date.

1.3.4.2.1. Notification shall include:

1.3.4.2.1.1. Aircraft tail number.

1.3.4.2.1.2. Date of last wash.

1.3.4.2.1.3. Reason for overdue condition.

1.3.4.2.1.4. Corrective action taken to prevent further occurrences.

1.3.4.2.2. MAJCOM Corrosion Program Manager ensures routing of waiver request to System Program Office (SPO) engineer and the Air Force Corrosion Prevention Control Office. The SPO engineer has final approval authority for waiver requests IAW T.O. 1-1-691 Chapter 3.

1.3.4.2.3. When aircraft are in severe corrosion environments, report overdue Clear Water Rinse (CWR) to AFGSC/A4M within seven calendar days after the 30th day of the CWR requirement IAW T.O. 1-1-691 Chapter 3.

1.3.5. Appoint, in writing, a Wing Corrosion Manager.

1.3.5.1. Wing Corrosion Manager will be a 2A7XX AFSC

1.3.6. **(Added-DYESSAFB)** Ensure Quality Assurance (QA) adequately evaluates corrosion control programs through inspection and maintenance follow-up evaluations. QA will evaluate a minimum of 10 percent of all aircraft washes.

**1.4. Wing Corrosion Manager Responsibilities.** Wing Corrosion Manager will:

1.4.1. Serve as the focal point for all aircraft and SE cleaning, corrosion and organic coating related information and taskings.

1.4.2. Serve as the point of contact for corrosion prevention and control between the MXG and AFGSC/A4M.

1.4.3. Ensure creation of a supplement to this instruction, to include, but not limited to the following:

1.4.3.1. Numbered Air Force (NAF), Wing, Operations Group (OG), Operational Test (OT) and Bomb Squadron commander aircraft designation and marking requirements of identified aircraft.

1.4.3.1. **(DYESSAFB)** Commander flagships will have a 2-inch white shadow on the “DY” unit designator. The “DY” is shadowed with the shadows facing the aft side of the letters (**Attachment 2**, **Figure 2**.).

1.4.3.1.1. **(Added-DYESSAFB)** 7th Bomb Wing commander's designation is "7 BW/CC".

1.4.3.1.2. **(Added-DYESSAFB)** 7th Operations Group commander's designation is "7 OG/CC".

1.4.3.1.3. **(Added-DYESSAFB)** 9th Bomb squadron commander's designation is "9 BS/CC.

1.4.3.1.4. **(Added-DYESSAFB)** 28th Bomb squadron commander's designation is "28 BS/CC".

1.4.3.1.5. **(Added-DYESSAFB)** 77th Weapons squadron commander's designation is "77 WPS" (Figure 13).

1.4.3.1.6. **(Added-DYESSAFB)** 337th Test and Evaluation squadron commanders' designation shall be "53 WG/337 CC" (**Figure 15**.).

1.4.3.2. Local unit marking requirements (e.g. tail stripes, crew names, nose art, etc.).

1.4.3.2.1. **(Added-DYESSAFB)** Commander's designations will be placed 3 1/2-inches below the "DY" patch and centered over the tail number.

1.4.3.2.2. **(Added-DYESSAFB)** Command aircraft (highlighted/shadowed) tail numbers will measure 10- inches in height and will be placed 4-inches below the specific commander's designation decal. On the left side of the vertical stabilizer, place the extreme right edge of the tail number 20-inches to the left of an invisible vertical line drawn down from the extreme right edge of the "DY" decal. On the right side of the vertical stabilizer, place the extreme left edge of the tail number 20-inches to the left of an invisible line drawn down from the extreme left edge of the "DY" decal and the center of the command patch.

1.4.3.2.3. **(Added-DYESSAFB)** For non-command aircraft (non-highlighted/non-shadowed), tail numbers will measure 15-inches tall and be placed 7-inches below the "DY" decal. Place the extreme right edge of the tail number 20-inches to the left of an invisible vertical line drawn down from the extreme right edge of the "DY" decal and the center of the command patch. On the left side of the vertical stabilizer, the extreme right edge of the "Y" shall be in line vertically with the bottom tip of the AFGSC patch. On the right side of the vertical stabilizer, the extreme left edge of the "D" shall be in line vertically with the bottom tip of the AFGSC patch.

1.4.3.3. Aircraft paint identification placard size, shape, and information requirements.

1.4.4. Train wash crew supervisors in all aspects of aircraft wash and to develop and implement a wash crew supervisor training plan. Units may designate a 7-level technician to fulfill these training responsibilities. **(T-3).**

1.4.5. Ensure development and utilization of local aircraft wash checklists IAW T.O. 00-5-1.

1.4.6. Ensure personnel are trained prior to taking receipt/custody of wash rack facilities and equipment.

1.4.6.1. Base-specific training is required for all newly assigned personnel. Training from another base is not sufficient.

1.4.6.2. Ensure training is documented in appropriate training records.

1.4.7. Enforce the use of approved coating materials and cleaning compounds as determined by T.O.s, Qualified Products Listings (QPL), and Qualified Products Database (QPD).

1.4.8. Ensure coating system scoring and maintenance is accomplished IAW **paragraph 2.2.2**.

1.4.9. Maintain records of all approved requests for aircraft names, nose art, tail flashes, internal nose art, and score sheets of maintained aircraft, as well as appointment letters for the Wing Corrosion Manager and Wash Rack Facility Manager. Maintain full length color photographs of all approved aircraft names and nose art along with approval documentation.

1.4.10. Attend assigned weapon systems CPAB, AF/DoD corrosion conferences, field surveys, and other Environmental/Technical conferences as specified by the Command Aircraft Corrosion Prevention and Control Program Manager.

**1.5. Maintenance Supervision Responsibilities.** Maintenance Supervision will:

1.5.1. Ensure owning activities wash and clean their aircraft and SE, as outlined in specific T.O.s. **(T-3).**

1.5.1.1. Ensure Wing Corrosion Manager and/or Aircraft Structural Maintenance (ASM) personnel assist the owning activities in their corrosion prevention efforts by accomplishing scheduled corrosion inspections on aircraft, support, and test equipment. **(T-3).**

1.5.2. Ensure wash crew supervisors are experienced/qualified 7-levels or above. **(T-3).** Wash crew supervisors will be trained by the Wing Corrosion Manager or a qualified 7-level ASM technician. **(T-3).** Training shall be documented in the wash crew supervisor's training record.

1.5.3. Ensure trained wash crew supervisors are present throughout the duration of aircraft washes. **(T-3).**

1.5.4. Provide a task trained and qualified aircraft wash crew. Ensure availability of Personal Protective Equipment (PPE) within the work center. When possible, the crew should include a dedicated crew chief or assistant dedicated crew chief.

**1.6. ASM/Low Observable Aircraft Structural Maintenance (LOASM) Section Chief Responsibilities.** ASM/LOASM Section Chief will:

1.6.1. Ensure no other maintenance is accomplished on aircraft, equipment, or within environmentally controlled/cordoned off areas when hazardous/toxic materials are in use that requires the use of specialized PPE. In the event specialized respiratory protection equipment is required, personnel will be properly fitted for the equipment and trained in its use IAW AFI 48-137, *Respiratory Protection Program.*

1.6.2. Forecast funding to attend and participate in applicable CPABs and other corrosion/structural related programs/meetings.

1.6.3. Submit CPAB agenda items to the Aircraft Corrosion Prevention and Control Program Manager in HQ AFGSC/A4M.

1.6.4. Appoint, in writing, a Wash Rack Facility Manager.

**1.7. Wash Rack Facility Manager Responsibilities.** Wash Rack Facility Manager will:

1.7.1. Ensure the required number and size of fire extinguishers are available and serviceable IAW weapon system specific technical data and local requirements.

1.7.2. Ensure grounding points are inspected and approved IAW T.O. 00-25-172.

1.7.3. Ensure fall protection equipment is available, used, and maintained IAW DAFMAN 91-203, *Air Force Occupational Safety, Fire, and Health Standards,* to allow coverage of all surface areas of aircraft during washing operations.

1.7.4. Ensure aircraft wash rack has cleaners identified in weapon system specific technical data. When cleaning products are not listed in weapon system specific technical data, ensure at least two types of approved cleaners IAW T.O. 1-1-691 are properly used, to include proper mix ratio and the correct cleaner for each area cleaned.

1.7.5. Ensure wash rack facility and surrounding area is kept clean and properly maintained.

1.7.6. Procure PPE used during wash process. Maintains wash rack facilities and equipment in serviceable condition (i.e., water hoses, pumps, air hoses, powered wash equipment, SE, PPE, etc.). This may not apply to units utilizing wash contracts.

**1.8. Wash Crew Supervisor Responsibilities.** Wash Crew Supervisor will:

1.8.1. Provide safety briefings explaining hazards associated with wash rack operations.

1.8.2. Ensure aircraft wash crews are task trained and qualified.

1.8.3. Ensure proper safety equipment, PPE, cleaning materials and fall protection lifeline cables (installed when required) are properly maintained IAW manufacturer's instructions and DAFMAN 91-203.

1.8.4. Document the requirement for wash, sign the wash completion, and document the lubrication requirement in the Air Force Technical Order (AFTO) Form 781A, *Maintenance Discrepancy and Work Document,* or other electronic form of documentation.

1.8.5. Ensure all wash rack equipment is inspected for serviceability (e.g., water hoses, pumps, air hoses, powered wash equipment, support equipment, etc.).

1.8.6. Ensure wash rack facility, surrounding area and equipment is clean and equipment is properly stored before and after use.

1.8.7. **(Added-DYESSAFB)** Notify the 7th Equipment Maintenance Squadron (7 EMS) corrosion section to receive wash supplies and safety briefing prior to beginning an aircraft wash. The 7th Maintenance Group (7 MXG) will use their own wash gear but require supplies. For assistance during non-duty hours, contact the 7 EMS production superintendent or the Aircraft Structures section with any questions regarding wash procedures. Aircraft washes occurring after duty hours must notify the 7 EMS production superintendent prior to beginning wash operations. The 317th Maintenance Group (317 MXG) will use their own supplies and wash gear.

1.8.8. **(Added-DYESSAFB)** Follow procedures listed in the Wash Rack Checklist (**Attachment 3**, Section 1 and Section 2).

1.8.9. **(Added-DYESSAFB)** Upon the completion of a wash and before the aircraft is removed from the wash facility, the 7 EMS corrosion section must be notified so that they may conduct a paint score of the aircraft and inspect it for corrosion and cleanliness. A Work Center Event (WCE) will be entered in Integrated Maintenance Data System (IMDS) for the 7 EMS corrosion section signifying that the aircraft requires corrosion inspection(s) and paint scoring. Corrosion will utilize a locally developed scoring method to score aircraft (**Attachment 4**). The score will be updated after each wash or semi-annually, depending upon the aircraft wash schedule.

# Chapter 2

# AIRCRAFT CORROSION PREVENTION AND CONTROL PROGRAM

## 2.1. General Policy.

2.1.1. Maintain aircraft paint scheme, markings, and decals IAW T.O.s 1-1-8, 1-1-691, 1-1-694 and this instruction.

2.1.1.1. Damage markings on Low Observable Aircraft is authorized IAW T.O. 1-1-694 Chapter 4 to easily identify damage and track defects.

2.1.2. Do not apply unauthorized decals or markings to aircraft. All deviations must be approved by the designated waiver authority and final packages will be sent to AFGSC/A4M for filing. Include digital photographs using criteria in **paragraph 2.2.4.23**.

2.1.3. Aircraft transferring from other commands should comply with AFGSC instructions within 90 days after transfer. **(T-3).**

2.1.4. Appearance Standards. Maintain aircraft paint, markings, and corrosion preventive coatings in a manner that will enhance the overall appearance and provide for the best corrosion protection.

2.1.4.1. Markings, warnings, and decals shall be legible and distinct.

2.1.5. Repainting aircraft, communications-electronic systems, or support equipment solely for beautification purposes is not authorized. **(T-3).**

2.1.6. CPAB requirements for aircraft units will be fulfilled through the annual aircraft/helicopter CPABs.

2.1.7. Apply soil barrier, as needed, to the tail boom areas affected by engine exhaust on the UH-1N and MH-139 helicopters. **(T-3).**

2.1.8. Aircraft coatings. Repair deteriorated aircraft coatings and areas of corrosion when discovered in order to provide continued surface protection. For metals unable to receive a primer, protect with an approved coating such as corrosion preventative compound.

## 2.2. Aerospace Vehicle Coating and Marking Requirements.

2.2.1. This section provides guidance for applying command approved, non-United States Air Force (USAF) standard aircraft coatings and markings as authorized in T.O. 1-1-8. Paint schemes/configurations and USAF standard aircraft markings will be applied in accordance with T.O. 1-1-8 and the applicable aircraft technical order.

2.2.2. Coating System Scoring and Maintenance. All AFGSC units, except for the 509 BW B-2 fleet, are required to score aircraft coating systems every 6 months, or as designated by aircraft specific technical guidance, to determine when restoration or touch-up is required. Aircraft painting will be scheduled on a “worst is first” basis to maintain coating system integrity. Supervisors will use rating to determine corrosion treatment/paint scheduling priority. Units are required to adopt maintenance-painting techniques IAW T.O. 1-1-8 to maintain aircraft corrosion protection between overcoats.

2.2.2.1. **(Added-DYESSAFB)** Each B-1B aircraft scheduled for touchup paint will be scheduled for seven (7) days (4 days sanding, touchup coatings, leading edge tape application and external markings; 3 days for paint cure). This does not include tail flash work. If aircraft paint coating condition score is 13 or above, additional paint time may be required. At a minimum, the following items will be removed prior to the aircraft entering the paint hangar unless re-paint of these items is not warranted: forward map case, trash receptacle, thermal flash protection shield containers, sustenance storage containers, maintenance grate, and central equipment bay door. (**NOTE:** TO 1B-1B-6WC-6, *Refurbish Inspection Workcards*, identifies Proud Flyer refurbishment requirements).

2.2.2.2. **(Added-DYESSAFB)** When prioritizing the schedule for aircraft and/or Support Equipment (SE) painting, the Aircraft Structural Maintenance (ASM) Pro Super, AGE Corrosion Control NCOIC, Maintenance Managers and an aircraft owning squadron representative (i.e., production superintendent must attend the shared resources meeting to coordinate and schedule paints based on date last painted, coating score and mission requirements).

2.2.3. Paint scheme. UH-1N aircraft will use a camouflage paint scheme on main airframes. MH-139 aircraft will maintain FED STD 36118 Grey Type IV paint scheme on main airframes.

2.2.4. Aircraft markings will be applied to aircraft as specifically authorized by this instruction, T.O. 1-1-8, and the applicable aircraft technical orders. LO aircraft markings not currently approved require a waiver from HQ AFGSC/A4M with System Program Directorate (SPD) approval due to strict survivability analysis requirements. Aircraft inputs to depot will be marked IAW with Air Force directives and this instruction only, unless otherwise approved by HQ AFGSC/A4M. HQ AFGSC/A4M is the point of contact for B-1, B-2, B-52, E-4B, UH-1N, MH-139, and T-38A aircraft painting and markings. All aircraft markings and basic paint schemes will be maintained intact, legible and distinct in color. Command standardization of markings by Mission Design Series (MDS) is of primary concern. AFGSC specific requirements are located within **Attachment 2**.

2.2.4.1. Stenciling. Markings may be applied using stencils or decals. Refer to T.O. 1-1-8 and T.O. 1H-1(U)N-3, *Structural Repair, USAF Series TH-1H and UH-1N Helicopters*, if applicable, to determine the compatibility of stenciling paints, paint finishes and decal applications.

2.2.4.2. Command Insignia. The application of the command insignia on aircraft is mandatory. Size and location of command insignias by MDS are specified in **Attachment 2**.

2.2.4.3. Organizational Insignia (Wing). The application of wing insignia is mandatory. Insignia will be applied IAW **Attachment 2**. **Note:** Wing Insignia is optional for UH-1N and MH-139 aircraft. If used, Wing insignias will be the same color scheme as the command insignia.

2.2.4.3.1. **(Added-DYESSAFB)** The 7 BW/CC's aircraft insignia shall consist of both squadron patches measuring 12-inches tall. They shall be placed side-to-side with the 9 BS patch forward of the 28 BS patch, on the left side of aircraft, centered 6-inches below the Weapons Systems Officer (WSO) window (**Figure 10**.).

2.2.4.4. Organizational Insignia (Squadron). Squadron insignia may be applied to aircraft at the wing's option IAW **Attachment 2**. **Note:** Squadron Insignia is optional for UH-1N and MH-139 aircraft. If used, Squadron insignias will be the same color scheme as the command insignia.

2.2.4.5. Distinctive Unit Aircraft Identification Marking. The application of the unit designator is mandatory for AFGSC aircraft unless otherwise directed. HQ AFGSC/A4M is the OPR for the assignment of unit designators. The primary factor used to determine appointment of unit designators is the aircraft/unit assignment location. T.O. 1-1-8, or the applicable aircraft technical guidance, will provide color restrictions and location for the unit designator. The unit designator will be applied in accordance with guidelines in **Attachment 2** and **Attachment 3**.

2.2.4.6. Tail Stripe/Art. Tail stripes/art are applied as a wing option, used to identify aircraft operation squadrons and display unit heritage. The Wing Commander must approve all tail stripe/art designs. Each operations squadron may have a tail stripe/art unique to that squadron. The use of the same tail stripe/art by two or more squadrons within a wing is not permitted. The stripe may be any color or pattern and may contain a logo. Photographs of locally approved tail stripe designs must be provided to HQ AFGSC/A4M for review and file. Units will not repaint tail flashes/stripes during deployed operations. See **Attachment 2. Note:** MH-139 does not have an approved tail stripe/art location.

2.2.4.6.1. Tail stripes/art previously approved are authorized, permitting designs meet criteria outlined in **paragraph 2.3.6**.

2.2.4.6.1.1. **(Added-DYESSAFB)** Tail stripes are accomplished by using two different methods: painting and vinyl. All checkered tail stripes use 220 vinyl Gerber tape and 1/2-inch overlapping edge sealer applied to prevent peeling during flight.

2.2.4.6.1.2. **(Added-DYESSAFB)** The 7 BW tail stripe base color will be sunflower yellow vinyl (220-20) with checkered (6-inch squares) black GCS-12 printed. It will be applied as one solid piece for each side with a front vinyl piece to connect both sides. The 9 BS and 28 BS patches will be applied to the aircraft.

2.2.4.6.1.3. **(Added-DYESSAFB)** The 9 BS tail stripe base color will be painted flat black (37038). The "bat" symbol will be stenciled with white paint (17925) and centered in the middle of the black stripe.

2.2.4.6.1.4. **(Added-DYESSAFB)** The 28 BS tail stripe base color will be intense blue (ECS-47) and matte white (220-20) checkered (6-inch squares) board tape.

2.2.4.6.1.5. **(Added-DYESSAFB)** The 337 TES tail stripe base color will be ruby red (GCS-053) and dark gray vinyl (220-41).

2.2.4.6.1.6. **(Added-DYESSAFB)** The 77 WPS tail stripe base color will be sunflower yellow vinyl (220-25) with checkered (6-inch squares) black tape (GCP-812).

2.2.4.6.1.7. **(Added-DYESSAFB)** The 489 BG tail stripe base color will be sunflower yellow vinyl (220-25) with ruby red (GCS-53) checkered (6-inch squares) board tape. The 489 BG patch will be applied to the aircraft.

2.2.4.6.1.8. **(Added-DYESSAFB)** The 345 BS tail stripe base color will be matte white vinyl (220-25) with ruby red (GCS-53) and black (GCS-12) checkered (6-inch squares) board tape with "CROOKS" staying white. The 489 BG patch will be applied to the aircraft.

2.2.4.7. Tail Markings (UH-1N applicable): See **Attachment 2.**

2.2.4.8. Aircrew and Crew Chief Names (Optional). Aircrew/Crew Chief names may be applied to all command aircraft**.** Crew Chief/assistant(s) names may be applied to all aircraft assigned to units with an established Dedicated Crew Chief (DCC) program. All names must be removed IAW T.O. 1-1-8 prior to deployment from home station in direct combat zones or when participating in contingencies that may subject aircraft to hostile fire abroad. Application of nicknames, punctuation, and/or call signs is not permitted. All aircraft in the wing will be standard with exception of the designated Commander's aircraft, which may have different lettering but will not exceed established height requirements (N/A for B-2). UH-1N lettering will not exceed 3" height. A background block for pilot/Crew Chief names may be used. The block should be in contrasting color to the section of the aircraft where applied. MH-139 does not currently have approved lettering. For B-1 and B-52 aircraft, the Wing Commander may approve a locally designed name block to promote unit pride. Photographs must be provided to HQ AFGSC/A4M for review and file. See **Attachment 2** for approved locally designed name blocks.

2.2.4.8.1. **(Added-DYESSAFB)** Aircraft commander, pilot/co-pilot, and/or WSO names will be stenciled with "brush script" lettering. The aircraft commander's name will be placed on the horizontal seam 2-inches below the left canopy and along an imaginary vertical reference line starting at the aft of the canopy edge. Pilot/co-pilot names will be applied on the right in the same manner. WSO names will be applied 1-inch below the aircraft commander and the pilot/co-pilot names.

2.2.4.8.2. **(Added-DYESSAFB)** Aircraft commander, pilot/co-pilot, WSO, and dedicated crew chief (DCC) and assistant dedicated crew chief (ADCC) names will be manufactured and applied by ASM personnel. Squadrons will request replacement of only two names on the nose landing gear door by initiating a 122 snapshot through IMDS and sending the request to the corrosion section (**Figure 12**.).

2.2.4.8.3. **(Added-DYESSAFB)** Squadrons will not arbitrarily change the painted aircrew or DCC/ADCC names outside of the established 30-/36-month schedule. Aircraft which require new aircrew or DCC/ADCC names applied or changed will be hangered inside the corrosion facility. This requirement is to prevent contamination of other hangars due to paint and stencil materials being sanded off the aircraft. The left nose gear door (containing DCC/ADCC names) can be removed and routed to the corrosion facility.

2.2.4.9. Commander's Aircraft Markings. Commander's aircraft referred to in this instruction are those designated as NAF, Wing, or OG commanders of flying squadrons and OT aircraft. The NAF Commander may select one wing within the command to have an aircraft specifically marked. Wing Commanders may select one aircraft per MDS to apply commander type markings. All other commanders are authorized only one aircraft each to be marked with standardized commander type markings. Standard insignias and markings will not be altered in location, dimension, or configuration to accommodate commander's aircraft markings. The following are markings authorized for use on commander's aircraft:

2.2.4.10. Wing and/or NAF insignias. The wing and/or NAF insignias will be applied on the right forward fuselage and a collage of assigned flight/operations squadron insignias will be applied on the left forward fuselage.

2.2.4.11. Highlighting of unit designator and bomb wing designator. All highlighting will be done in contrasting gray, black or white if it meets primary basecoat gloss requirements (i.e., gloss, camouflage or gunship). All unit designators and serial numbers will remain on vertical stabilizers or as depicted in applicable technical orders.

2.2.4.12. Wing Commanders must approve the markings. Final packages will be sent to AFGSC/A4M for filing. Include digital photographs using criteria in **paragraph 2.2.4.23**.

2.2.4.13. Nose Numbers (optional). If used, aircraft nose numbers shall be in block or Helvetica letters, not to exceed five digits. Specific location and size for each different type aircraft is listed in **Attachment 2**. The paint material(s) used to apply nose numbers shall have the same gloss requirement as the base aircraft coating.

2.2.4.14. Bomber Combat Marking. Designated bomber aircraft with a successful weapons release in combat operations are authorized to display a conventional bomb and/or Joint Air-to Surface Stand-Off Missile (JASSM) silhouette on that aircraft. These markings will be applied in contrasting shades that conform to the basic aircraft camouflage requirements. B-1, B-2 and B-52 combat marking configurations are located within **Attachment 2**.

2.2.4.15. Nose Art. For purposes of clarification, "nose art" shall be the term used to identify specialized artwork applied to any area of the aircraft. Wing Commanders may designate one aircraft within the wing to apply the "Let's Roll" design. The aircraft selected is at the Wing Commander's discretion. Waivers for additional aircraft will not be considered. B-1 and B-52 art is authorized on the left side of the aircraft only. B-2 nose art is authorized on the inner mold line of the left main landing gear door. E-4B nose art is authorized on the interior of the crew entry area in conjunction with the crew chief placard and/or on the exterior, left side of the aircraft only. UH-1N nose art is authorized on the Upper Hinged Nose Door of the aircraft only and art is limited to size of the door. Authorized nose art is limited to 3 feet by 3 feet in size. All nose art information shall be incorporated in each base's supplement to this instruction. Nose art shall meet the following criteria:

2.2.4.15.1. Be distinctive, symbolic, and designed in good taste.

2.2.4.15.2. Enhance unit pride.

2.2.4.15.3. Comply with equal opportunity policies.

2.2.4.15.4. Match gloss requirements of the basic paint scheme and utilize MDS approved paint or stencil material.

2.2.4.15.5. All designs will be reviewed per **paragraph 1.2.1** and approved by WG/CC. Once approved by the WG/CC, send final packages to AFGSC/A4M for filing. Include digital photographs using criteria in **paragraph 2.2.4.23**. AFGSC/A4M will ensure packages are forwarded to AFGSC/HO and WG Historian.

2.2.4.15.6. Nose art previously approved is authorized, permitting designs meet criteria outlined in **paragraph 2.2.4.15** thru **paragraph 2.2.4.15.5**.

2.2.4.16. Unique Unit Markings.

2.2.4.16.1. Anniversary Markings. When applied, anniversary markings will not interfere with required aircraft markings and must be removed immediately after the anniversary period (1-year maximum). Waivers are not required. Wing Commanders must approve the markings, and digital photographs must be provided to AFGSC/A4M for review and file.

2.2.4.16.2. State flags and logos other than anniversary type are not considered unit unique markings.

2.2.4.17. Aircraft Naming. Route requests to name AFGSC aircraft through wing Public Affairs (PA) to AFGSC/A4M. AFGSC/A4M will coordinate with AFGSC/PA before submitting to AFGSC/A4 for approval. This includes markings previously considered unit unique and are community related/appreciation types such as “Spirit of,” “City of” and “State of.” Naming aircraft is a tradition designed to commemorate or honor individuals, geographic locations, or events either for the support provided by the Air Force on a long-term basis, or because of its significance to Air Force history or heritage. Recommendations must include a proposed name, suggested aircraft and tail number, and detailed justification for the proposed name. Contact wing Public Affairs office for details.

2.2.4.18. Aircraft Travel Pods. Travel pods will be painted in gloss paint the same color as the aircraft with no additional markings. Travel pods designated for commanders may contain the position and name of the individual and appropriate insignia. Lettering may be of any color and style, but shall not exceed 6” height. Commander’s travel pod paint scheme, final marking sizes, and placements will be approved by the WG/CC and documented within local unit operating instructions. **(T-3).**

2.2.4.19. Paint Identification Placard. The paint identification block is a mandatory marking. The block may be of a unique design, or state outline, but must not exceed 6” by 6” size and will match the color of other markings on the aircraft. Waiver requests are not required for this item. However, WG/CC approval is required, and photographs of the design must be provided to HQ AFGSC/A4M for review and file (not applicable for B-2).

2.2.4.20. Competition Aircraft. Units participating in competitions will follow the guidelines established in competition rules for aircraft appearance. Competitions shall be considered "come as you are" and no waivers will be granted. "Come as you are" is defined as no special effort, painting, or additional markings applied to enhance or improve the overall appearance of the aircraft. This includes polishing of titanium, commander markings, etc.

2.2.4.21. Aircraft Transfer. The following markings must be removed prior to formal transfer of aircraft to other units or MAJCOMs (aircraft retiring to the Aerospace Maintenance and Regeneration Group, need not have any markings removed).

2.2.4.21.1. Organizational insignias.

2.2.4.21.2. Distinctive unit aircraft identification.

2.2.4.21.3. Tail stripe.

2.2.4.21.4. Aircrew and Crew Chief names.

2.2.4.21.5. Unit-unique markings.

2.2.4.21.6. Nose art may be retained if gaining unit agrees.

2.2.4.22. Waivers. Wing Commanders must submit waiver requests for non-standard markings to HQ AFGSC/A4M for approval/disapproval. Waivers that are in violation of aircraft technical orders will not be accepted. Waiver requests must include the following:

2.2.4.22.1. Clear statement of present procedure/marking.

2.2.4.22.2. Clear statement of proposed change.

2.2.4.22.3. Justification to include historical significance, if applicable.

2.2.4.22.4. Photographs: Two high-quality digital color photographs, one of present marking and one of requested change.

2.2.4.23. Photo Requirements. All photo requirements may be met by a high-quality digital photograph. All AFGSC units must submit one full length (landscape orientation) of the commander's aircraft each time a marking change occurs to HQ AFGSC/A4M for review and file. Units will provide photos of unique markings for all local option changes authorized by this instruction (e.g., tail stripe/name block design and/or color changes, paint data placard, nose art, etc.) to HQ AFGSC/A4M for review and file.

2.2.4.24. Total Force Integration (TFI). For TFI locations under the classic association, coordination is required from both Wing Commanders to identify specific aircraft for unit identification and/or organizational insignia changes. These proposed changes are in support of ownership, pride and esprit de corps between active duty, reserve, and guard partnering wings. Once insignia change is approved by the owning WG/CC, the supporting documents will be routed to HQ AFGSC/A4M for validation IAW T.O. 1-1A-8.

**2.3. AGE Tone-Down Procedures.**

2.3.1. When feasible, equipment will be completely stripped and properly prepared IAW T.O. 1-1-8 and T.O. 35-1-3, *Corrosion Prevention and Control, Cleaning, Painting, and Marking of USAF Support Equipment (SE)* before applying polyurethane coatings.

2.3.2. Minimum reflectorizing requirements will be IAW T.O. 35-1-3. Black subdued reflectorized tape will be used in lieu of white when left optional by T.O. 35-1-3.

2.3.3. Dedicated squadron identification markings will not exceed a 2" by 6" area below two field numbers if the equipment area permits.

2.3.4. Locally devised field numbers will be black.

2.3.5. Interior areas of AGE exposed during operation will be toned down to match exterior painted surfaces. **(T-3).**

2.3.6. AGE arriving on base and requiring tone down will be scheduled for paint within 90 days of receipt, and completed within 24 months. **(T-3).**

**2.4. (Added-DYESSAFB) Aerospace Ground Equipment (AGE) Flight Chief Responsibilities.**

2.4.1. **(Added-DYESSAFB)** AGE will be scheduled for immediate repaint if 75 percent or more of the coating has failed (Category 4). IAW TO 35-1-3, *Corrosion Prevention and Control, Cleaning, Painting, and Marking of USAF Support Equipment (SE)*, painting will be accomplished for corrosion protection only, not for beautification purposes.

**2.5. (Added-DYESSAFB) AGE Corrosion NCOIC Responsibilities.**

2.5.1. **(Added-DYESSAFB)** Updates changes to equipment rating in their localized database. Personnel in the corrosion section will have access to AGE corrosion information to enable review fleet status.

2.5.2. **(Added-DYESSAFB)** Ensures AGE corrosion inspections are be rated IAW TO 35-1-3.

2.5.3. **(Added-DYESSAFB)** Updates reports on the status of age prior to the last duty day of each month. This will be accomplished by listing the number of units in each of the above ratings. The report will be sent to the AGE production section or designated representative.

2.5.4. **(Added-DYESSAFB)** Attends the monthly shared resources meeting (as applicable). The AGE corrosion control NCOIC and munitions control NCOIC will ensure support equipment is scheduled to be prepared for the corrosion control section at least one week in advance.

2.5.5. **(Added-DYESSAFB)** DPAS or Autonomic Logistics Information System (ALIS) is used to schedule and document AGE painting.

2.5.6. **(Added-DYESSAFB)** Coordinates with the AGE scheduler to schedule equipment for paint. All equipment that is scheduled for paint will have a discrepancy annotated in the applicable forms. A WCE for the corrosion section will be entered in IMDS signifying that the unit requires paint. The AGE flight production superintendent or designated representative will coordinate delivery of equipment scheduled for paint. A completed AFTO 350 Tag, *Repairable Item Processing*, will accompany all equipment items delivered to AGE corrosion control. Prior to delivery, AGE flight will remove bumper pads, power cables, heater ducts, external forms bags, etc., that do not require painting.

2.5.6.1. **(Added-DYESSAFB)** All other agencies will coordinate support equipment through the corrosion control section for scheduling requirements. A WCE for the corrosion control section will be entered in IMDS signifying that the unit requires painting.

2.5.7. **(Added-DYESSAFB)** Ensures maintenance, servicing and inspection activity personnel are provided an oriented of the corrosion control section.

2.5.8. **(Added-DYESSAFB)** Ensures proper corrosion treatment is accomplished at scheduled inspections. Units having bare metal or corroded areas will be treated and painted IAW appropriate technical data during maintenance actions.

2.5.9. **(Added-DYESSAFB)** Ensures powered and non-powered AGE is cleaned, thoroughly inspected and touched-up as necessary during each periodic inspection with the appropriate and approved materials.

2.5.10. **(Added-DYESSAFB)** Ensures new components that do not meet the required MIL-spec color will be processed and painted prior to installation.

2.5.11. **(Added-DYESSAFB)** Enforces the proper use of approved cleaning compounds IAW TO 35-1-3 and the Qualified Product Listing (QPL) or Qualified Product Database (QPD). Obtains QPLs from the wing corrosion control manager and uses them IAW applicable technical data to verify all compounds on-hand are authorized for use on designated equipment.

2.5.12. **(Added-DYESSAFB)** Schedules work beyond AGE work center capability to the appropriate fabrication work center.

2.5.13. **(Added-DYESSAFB)** Ensures tone-down procedures are followed IAW AFGSCI 21-105.

2.5.14. **(Added-DYESSAFB)** Will schedule corrosion inspections for corrosion control personnel. These inspections will be conducted within the AGE flight facility during other scheduled maintenance inspections, to ensure the scheduled inspections are all performed simultaneously to reduce equipment downtime. If AGE is unavailable or unmovable, the AGE corrosion control NCOIC will assist corrosion control section personnel in locating equipment.

## Chapter 3

## ICBM CORROSION PREVENTION AND CONTROL ROLES AND RESPONSIBILITIES

**3.1. HQ AFGSC/A4B Responsibilities.** HQ AFGSC/A4B is responsible for overall Intercontinental Ballistic Missile (ICBM) corrosion program management within the command. HQ AFGSC/A4B will appoint an ICBM Corrosion Prevention and Control Functional Manager. The functional manager will:

3.1.1. Manage AFGSC's Corrosion Prevention and Control Program for the ICBM weapon system.

3.1.2. Serve as the command's focal point for corrosion prevention and control issues, and coordinate all actions with staff directorates as appropriate.

3.1.3. Ensure adequate corrosion control training is available and current for all ICBM maintenance personnel.

3.1.4. Serve as the Command's interface with Air Education and Training Command (AETC) and the Air Force Institute of Technology (AFIT) for corrosion training course content changes.

3.1.5. Ensure Air Force Materiel Command conducts annual site surveys for ICBM facilities.

3.1.6. Represent Command at assigned weapon systems CPAB, AF/DoD corrosion conferences, field surveys, and other Environmental/Technical conferences as applicable to ICBMs.

**3.2. Wing Commander Responsibilities.** The WG/CC will:

3.2.1. Approve all Missile Alert Facility (MAF) mural requests. Ensure all requests have been routed through Corrosion Manager for tracking.

**3.3. Maintenance Group and 576 FLTS Commander Responsibilities.** The MXG/CC and 576 FLTS/CC are responsible for establishing and maintaining effective Corrosion Prevention and Control Programs. The MXG/CC and 576 FLTS/CC will:

3.3.1. Ensure adequate facilities, equipment, manpower, material and funding are available to support a sound corrosion prevention and control program.

**3.4. Unit Corrosion Prevention and Control Program Manager Responsibilities.** Unit Corrosion Prevention and Control Program Manager will:

3.4.1. Also be the Corrosion Shop Foreman for their respective units. (576 FLTS will utilize a Corrosion Shop Foreman or QAE only). **(T-3).**

3.4.2. Act as the Subject Matter Expert (SME) for all corrosion and coatings related information and taskings.

3.4.3. Serve as the point of contact for corrosion prevention and control between the unit and AFGSC/A4B.

3.4.4. Develop and publish local corrosion prevention training requirements, as required. **(T-3).**

3.4.5. Ensure military technicians, Department of Air Force civilians, and government contracted corrosion control technicians:

3.4.5.1. Inspect for and treat corrosion IAW technical orders using approved products.

3.4.5.2. Handle and dispose of hazardous materials/waste IAW local, state, and federal guidelines.

3.4.6. Ensure only authorized chemical cleaning materials and corrosion removal methods are used, and that Safety Data Sheets are available for each chemical used.

3.4.7. Ensure required equipment is obtained for an efficient and effective corrosion prevention and control program.

3.4.8. Enforce the use of approved coating materials and cleaning compounds as determined by TOs, QPL, and QPD.

3.4.9. Attend annual CPAB, AF/DoD Corrosion Conferences, Foreman's Meeting and other Environmental/Technical conferences as specified by the ICBM Corrosion Prevention and Control Functional Manager.

3.4.10. Track MAF mural information.

3.4.11. Ensure personnel dispatched to Launch Facilities (LFs) and MAF to perform maintenance on missile systems will receive initial and annual ICBM corrosion prevention and control training IAW AFMAN 21-202, *Missile Maintenance Management*. ICBM corrosion control specialists are exempt from this requirement.

3.4.12. Ensure ICBM maintenance units use standardized AFGSC products for ICBM corrosion prevention and control training.

3.4.13. Ensure ICBM corrosion control specialists attend site-specific environmental compliance training as designated by the installation Environmental office. **(T-3).**

# Chapter 4

# ICBM CORROSION PREVENTION AND CONTROL PROGRAM

## 4.1. General Policy.

4.1.1. Only qualified government or contract corrosion control technicians will perform coating removal/application tasks when required by applicable technical guidance.

4.1.1.1. Repainting weapon system components and structures solely for beautification purposes is not authorized.

4.1.2. Missile Engineering must coordinate all Master Change Log activities and real property/real property installed equipment changes through the Unit Corrosion Prevention and Control Program Manager.

4.1.3. On-base LF trainers will be scheduled on a 3-year cycle for corrosion inspection/treatment using the applicable LF -6 work cards.

4.1.4. Federal, state, and local environmental regulations must be followed.

4.1.5. The Society for Protective Coatings (SSPC) and the National Association of Corrosion Engineers (NACE) offer various training courses for coating systems, application and inspection. As unit funds are available, corrosion program/functional managers, instructors and evaluators should attend SSPC or NACE course(s) that relate to their position and enhance their professional knowledge.

4.1.6. Utilization of corrosion specialists for anything other than weapon system maintenance requires authorization of the MXG/CC or 576 FLTS/CC. (T-3).

4.1.7. The ICBM Corrosion Prevention and Control Program consists of frequent cleaning, corrosion inspections and early detection, application of proper treatment materials/procedures, and maintenance painting. Maintenance painting is defined for field purposes as spot painting, sectionalized painting, and complete scuff sand and overcoat. The corrosion prevention and control program is divided into three components.

4.1.7.1. Prevention. Corrosion prevention must be a proactive part of the overall program. Use of proper materials and technical order procedures during maintenance activities are critical to program success.

4.1.7.1.1. Frequent cleaning has proven to be the most effective means of preventing corrosion.

4.1.7.1.2. All maintenance personnel must perform general housekeeping and cleaning as part of every maintenance activity, supervisors are required to enforce these standards.

4.1.7.2. Detection.

4.1.7.2.1. All maintenance personnel will perform corrosion inspections as part of routine maintenance activities.

4.1.7.2.2. Supervision at all levels must emphasize active participation by all technicians during scheduled periodic inspections as well as unscheduled maintenance.

4.1.7.2.3. Conduct inspections for corrosion in concealed/hard to access locations when opened for any other work.

4.1.7.2.4. Inspections will be documented in the appropriate MIS.

4.1.7.3. Treatment.

4.1.7.3.1. Units will maintain weapon system components IAW **Attachment 4**.

4.1.7.3.2. There are two general categories of treatment.

4.1.7.3.2.1. Remove and replace. Performed by maintenance technicians responsible for system maintenance when component is beyond repair or treatment in place is not possible or cost effective.

4.1.7.3.2.2. Treatment in place. Minor corrosion can be treated by maintenance technicians during routine maintenance as directed by weapon system specific T.O.s. Large-scale or more severe corrosion on installed equipment/components will be treated by corrosion specialists.

4.1.7.3.3. LF topside areas are extremely susceptible to environmental factors. The Corrosion Shop Foreman must exercise sound professional judgment when determining when to spot paint or completely recoat surfaces. Complete recoats are authorized for topside surfaces.

4.1.7.3.3.1. For areas other than LF topside, limit painting to the smallest practical area required to maintain a professional appearance and an effective coating system.

4.1.7.3.4. All coating systems used on weapon system components and structures, to include real property installed equipment and operational ground equipment, will comply with technical requirements in T.O.s 21M-LGM30F-101, *LGM30 Weapon System Corrosion Control and Treatment*, 1-1-8, 1-1-691, 35-1-3, or applicable equipment T.O.s.

4.1.7.3.5. Units will deplete existing stocks of paint prior to switching to color schemes specified in this instruction. All coatings purchased after implementation of this instruction will be IAW **Attachment 4**.

4.1.7.3.6. A detailed paint plan for living facilities or other above ground facilities at the MAF is not required.

4.1.7.3.7. Repainting missile system facilities/components, communications-electronic systems, or support equipment solely for beautification purposes is not authorized.

4.1.7.3.8. Do not paint equipment/components where the operational capability or designed function of an item would be impaired by paint. Corrosion specialists will not paint the following areas:

4.1.7.3.8.1. Fabric or plastic surfaces.

4.1.7.3.8.2. Bare or untreated concrete surfaces. **Exception:** warning lines, caution lines, designators, etc. required by technical orders.

4.1.7.3.8.3. Surfaces in contact with weather seals and gaskets. **Exception:** surfaces specifically directed by T.O. 21M-LGM30F-101 and T.O. 21M-2 LGM30F-112, *General Inspection and Repair Procedures ICBM Systems*.

4.1.7.3.8.4. Radio Frequency Interference (RFI) gaskets/shields and surfaces in contact with RFI gaskets/shields.

4.1.7.3.8.5. Machined surfaces of moving parts.

4.1.7.3.8.6. Identification plates and warning signs.

4.1.7.3.8.7. Lubrication devices and grease fittings.

4.1.7.3.8.8. Electrical terminal strips, ground straps, connectors, wires, and bus bars.

4.1.7.3.8.9. Blast valve assemblies, hydraulic and pneumatic tubing, fittings and controls.

4.1.7.3.8.10. Shock Isolator piston rods.

4.1.7.3.8.11. Abrasive (non-skid) surfaces/floor plates.

4.1.7.3.8.12. Asphalt shingles.

4.1.7.3.8.13. Equipment drawers.

4.1.7.3.8.14. Sound-proofed surfaces.

4.1.7.3.8.15. Wooden ladders.

**4.2. Murals.**

4.2.1. MAF Murals (optional). For purposes of clarification, “mural” shall be the term used to identify specialized artwork applied to a MAF. All mural information, including location and size limitations as applicable, shall be incorporated in each base’s supplement to this instruction. Murals shall meet the following criteria:

4.2.1.1. Be distinctive, symbolic, and designed in good taste.

4.2.1.2. Enhance unit pride.

4.2.1.3. Comply with equal opportunity policies.

4.2.1.4. All designs will be reviewed and approved by the WG/CC. **(T-3).**

KENYON K. BELL
Brigadier General, USAF
Director, Logistics and Engineering

**(DYESSAFB)**

SAMUEL A. FRIEND, Colonel, USAF
Deputy Commander, 7th Bomb Wing

**Attachment 1**

**GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION**

***References***

**(Added-DYESSAFB)** AFI 33-322, *Records Management and Information Governance Program,* 28 July 2021

AFPD 21-1, *Maintenance of Military Material,* 1 Aug 2018

AFPD 13-5, *Air Force Nuclear Enterprise,* 17 Jul 18

AFI 33-322, *Records Management and Information Governance Program,* 28 Jul 21

AFI 48-137, *Respiratory Protection Program*, 12 Sep 18

AFMAN 21-202, *Missile Maintenance Management*, 29 Aug 19

AFMAN 91-203, *Air Force Occupational Safety, Fire, and Health Standards,* 11 Dec 18

DAFI 21-101, *Aircraft and Equipment Maintenance Management,* 16 Jan 20

DAFMAN 32-1084, *Standard Facility Requirements,* 15 Jan 20

DAFMAN 90-161, *Publishing Processes and Procedures,* 15 April 2022

T.O. 00-5-1, *AF Technical Order System,* 25 Jan 21

T.O. 00-25-172, *Ground Servicing of Aircraft and Static Grounding/Bonding,* 21 Aug 21

T.O. 1-1-8, *Application and Removal of Organic Coatings, Aerospace and Non-Aerospace Equipment,* 15 Dec 20

T.O. 1-1-691, *Cleaning and Corrosion Prevention and Control, Aerospace and Non-aerospace Equipment,* 21 Aug 21

T.O. 1-1-694, *Application and Removal of Low Observable Coatings on Aerospace*, 6 May 19

T.O. 1B-2A-2-11GS-00-1, *Placards and Markings,* 10 Mar 21

T.O. 1H-1(U)N-3 *Structural Repair, USAF Series TH-1H and UH-1N Helicopters*, 1 Jun 20

T.O. 21M-LGM30F-101, *LGM30 Weapon System Corrosion Control and Treatment VAFB, Wings I, III, and V USAF SERIES LGM30G MISSILE,* 27 Jul 20

T.O. 21M-LGM30F-112, *General Inspection and Repair Procedures ICBM Systems,* 16 May 21

T.O. 35-1-3, *Corrosion Prevention and Control, Cleaning, Painting, and Marking of USAF Support Equipment (SE),* 10 Apr 21

**(Added-DYESSAFB)** TO 1-1-691, *Aircraft Weapons Systems Cleaning and Corrosion Control,* 22 July 2022

**(Added-DYESSAFB)** TO 1B-1B-6WC-6, *Refurbish Inspection Workcards,* 15 January 2023

**(Added-DYESSAFB)** TO 35-1-3, *Corrosion Prevention and Control, Cleaning, Painting, and Marking of USAF Support Equipment (SE),* 26 January 2022

UFC 4-211-02, *Aircraft Corrosion Control and Paint Facilities*, 1 Dec 12

***Prescribed Forms***

No forms are prescribed by this instruction

***Adopted Forms***

AF Form 847, *Recommendation for Change of Publication*

(**Added-DYESSAFB**) AFTO 350, *Repairable Item Processing Tag*

AFTO Form 781A, *Maintenance Discrepancy and Work Document*

(**Added-DYESSAFB**) DAF Form 847, *Recommendation for Change of Publication*

***Abbreviations and Acronyms***

**(Added-DYESSAFB) ADCC**—Assistant Dedicated Crew Chief

**(Added-DYESSAFB) ADO**—Assistant Director of Operations

**AETC**—Air Education and Training Command

**AFRC**—Air Force Reserve Command

**AFGSC**—Air Force Global Strike Command

**AFGSCI**—Air Force Global Strike Command Instruction

**AFI**—Air Force Instruction

**AFIT**—Air Force Institute of Technology

**AFMAN**—Air Force Manual

**AFPD**—Air Force Policy Directive

**AFRC**—Air Force Reserve Command

**AFRIMS**—Air Force Records Information Management System

**AFTO**—Air Force Technical Order

**AGE**—Aerospace Ground Equipment

**(Added-DYESSAFB) ALIS**—Autonomic Logistics Information System

**ANG**—Air National Guard

**ASM**—Aircraft Structural Maintenance

**(Added-DYESSAFB) BS**—Bomb Squadron

**CPAB**—Corrosion Prevention Advisory Board

**DAFI**—Department of the Air Force Instruction

**DAFMAN**—Department of the Air Force Manual

**DCC**—Dedicated Crew Chief

**DoD**—Department of Defense

**(Added-DYESSAFB) DPAS**—Defense Property Accountability System

**ESA**—Electrical Surge Arrestor

**(Added-DYESSAFB) FOD**—Foreign Object Damage

**(Added-DYESSAFB) GOV**—Government Owned Vehicle

**HQ**—Headquarters

**IAW**—In Accordance With

**ICBM**—Intercontinental Ballistic Missile

**(Added-DYESSAFB) IMDS**—Integrated Maintenance Database System

**JASSM**—Joint Air-to Surface Stand-Off Missile

**LCC**—Launch Control Center

**LF**—Launch Facility

**LO**—Low Observable

**LOASM**—Low Observable Aircraft Structural Maintenance

**MAF**—Missile Alert Facility

**MAJCOM**—Major Command

**MDS**—Mission Design Series

**MIS**—Maintenance Information System

**MXG**—Maintenance Group

**NACE**—National Association of Corrosion Engineers

**NAF**—Numbered Air Force

**(Added-DYESSAFB) NCOIC**—Noncommissioned Officer-in-Charge

**OG**—Operations Group

**OPR**—Office of Primary Responsibility

**OT**—Operational Test

**PA**—Public Affairs

**PPE**—Personal Protective Equipment

**(Added-DYESSAFB) QA**—Quality Assurance

**QAE**—Quality Assurance Evaluator

**QPD**—Qualified Products Database

**QPL**—Qualified Products Listings

**RDS**—Records Disposition Schedule

**RFI**—Radio Frequency Interference

**SE**—Support Equipment

**SME**—Subject Matter Expert

**SPD**—System Program Directorate

**SPO**—System Program Office

**SSPC**—Society for Protective Coatings

**TFI**—Total Force Integration

**T.O.**—Technical Order

**(Added-DYESSAFB) TO**—Technical Order

**URD**—Unit Reference Designator

**USAF**—United States Air Force

**(Added-DYESSAFB) WCE**—Work Center Event

**(Added-DYESSAFB) WSO**—Weapons Systems Officer

## Attachment 2

## AIRCRAFT MARKING SPECIFICATIONS

**A2.1. Aircraft Marking Specifications.** Due to the mission of the E-4B, no command markings will be applied to the aircraft.

**Table A2.1. B-1 Aircraft Marking Specifications.**

| B-1 | |
|---|---|
| **COMMAND INSIGNIA:** | 24" (subdued). Located 11" down from the tail stripe, top of patch 38" from tail leading edge and 36" from leading edge of rudder on both sides. |
| **ORGANIZATIONAL INSIGNIA:** | 24" (subdued). Located 6" below and centered on the OSO/DSO windows. |
| **UNIT IDENTIFIER:** | 30". Located 7" down from command insignia. On right side of tail, trailing edge of first letter will lay along a vertical line from center of sword in command insignia. On left side of tail, trailing edge of second letter in unit designator will lay along a vertical line from center of sword in command insignia. |
| **TAIL STRIPE:** | Wings will identify tail stripe specifications for authorized variants in supplements to this instruction. |
| **PILOT/AIRCREW/CREW CHIEF NAMES:** | Pilot/aircrew: Centered on forward escape hatch side window.<br>Crew chief/assistant: Left nose gear door, centered. |
| **NOSE NUMBERS:** | Last three/four digits of tail number, 3" height, on nose gear strut, both sides. |
| **COMBAT MARKINGS:** | Bomb markings will be located on the right portion of the left nose gear door exterior. These markings will consist of a 5" conventional bomb or JASSM silhouette applied in a row not to exceed 10 in each row. Once 10 silhouettes are achieved, a single conventional bomb or JASSM silhouette with the number a "10" will represent a row. |

**Table A2.2. B-2 Aircraft Marking Specifications.**

| B-2 | |
|---|---|
| **COMMAND INSIGNIA:** | 24” (subdued). Located on left side at FS 304.5 (center of crest), with crest bottom at WL 167.2. |
| **ORGANIZATIONAL (WING) INSIGNIA:** | 24” (subdued). Located on right side of aircraft at FS 304.5 (center of crest), crest bottom at WL 167.2. The WG/CC may approve an additional insignia to be installed 6” aft of organizational insignia on aircraft for esprit-de-corps in classically associated TFI units. |
| **DISTINCTIVE UNIT AIRCRAFT IDENTIFICATION MARKING:** | 16” (subdued). 30° negative slant. Top of unit designator to be 15” from top of main landing gear door, 50” from extreme aft end of door. |
| **TAIL STRIPE:** | The “tail stripes” will be 18’ long by 11.25” high from leading edge to trailing edge of main landing gear doors. |
| **AIRCRAFT NAME:** | (all measurements to top of letters)<br>“SPIRIT” located: 10.5” from top of main landing gear door, 57” from extreme forward of door in 8” block letters with a 30° negative slant.<br>“OF” located: 19.5” down from top of main landing gear door, centered underneath the “SPIRIT” in 6” block letters with a 30° negative slant.<br>“STATE/CITY” located: 26.5” from top of main landing gear door, centered beneath the “SPIRIT” in 8” block letters with a 30° negative slant. |
| **PILOT/AIRCREW/CREW CHIEF NAMES:** | Pilot/Aircrew/Crew Chief names are applied IAW TO 1B-2A-2-11GS. |
| **NOSE NUMBERS:** | Nose numbers are applied IAW TO 1B-2A-2-11GS. |
| **COMBAT MARKINGS:** | Units may apply a single bomb or JASSM silhouette, not to exceed 3” height, on the exterior of the nose landing gear door for each combat sortie flown with a successful weapons release. These silhouettes will contain the appropriate combat operation acronym, (e.g., “OEF”) embedded in the center of the marking. |

**Table A2.3. B-52 Aircraft Marking Specifications.**

| B-52 | |
|---|---|
| **COMMAND INSIGNIA:** | 24” (subdued). Insignia is applied to both sides of the vertical stabilizer. Top of insignia will be located 41.5” below tail stripe. Leading edge of insignia will be 62” from leading edge of tail on both sides of tail. Bottom point/tip of insignia will be located on center seam of main spar line. |
| **ORGANIZATIONAL INSIGNIA:** | 24” (subdued).<br>Right side: On forward fuselage, center of insignia at B.S. 218.7 and W.L.187.5.<br>Left side: On forward fuselage, center of insignia at B.S. 218.7 and W.L.187.5. |
| **DISTINCTIVE UNIT AIRCRAFT IDENTIFICATION MARKING:** | Use two-letter base designation **(Attachment 3)** in 42” black block letters.<br>Left side: Top of unit designator located 36” below box antenna. The trailing edge of the second letter is on a vertical line down from the tip of the command insignia, using the lower point of the center lightning bolt in the insignia as a line up reference.<br>Right side: Top of unit designator located 36” below box antenna. The trailing edge of the first letter is on a vertical line down from the tip of the command insignia, using the lower point of the center lightning bolt in the insignia as a line up reference. |
| **DISTINCTIVE UNIT AIRCRAFT IDENTIFICATION MARKINGS ON COMMANDER’S AIRCRAFT:** | Use two-letter base designation **(Attachment 3)** in 42” black block letters. When used in conjunction with the bomb wing designator the location will be as follows:<br>Left Side: Top of unit designator located 20” below box antenna. The trailing edge of the second letter is on a vertical line down from the tip of the command insignia, using the lower point of the center lightning bolt in the insignia as a line up reference.<br>Right Side: Top of unit designator located 20” below box antenna. The trailing edge of the second letter is on a vertical line down from the tip of the command insignia, using the lower point of the center lightning bolt in the insignia as a line up reference. |
| **DISTINCTIVE UNIT AIRCRAFT IDENTIFICATION MARKINGS ON BOMB WING COMMANDER’S AIRCRAFT:** | Use two-letter base designation **(Attachment 3)** in 18” black block letters.<br>Left and right side: Top of bomb wing designator will be located 6” below unit designator and aligned with serial numbers. |

| | |
|---|---|
| **TAIL STRIPE:** | The tail stripe will be applied at the upper portion of the vertical stabilizer and must be in the form of a straight horizontal stripe. The overall height will not exceed 15”. |
| **PILOT/AIRCREW/CREW CHIEF NAMES:** | Pilot/aircrew: Centered under pilot’s window.<br>Crew Chief/assistant: Block is 15” by 36”. Block is located on BS 261.00 and WL 139.00.<br>2 BW Crew chief/assistant: Block is the shape of Louisiana and 30” by 24”. Forward bottom edge of block is aligned with BS 258.70 and WL 139.<br>5 BW Crew Chief/assistant: Block is the shape of North Dakota and 15” by 35.547”. Forward bottom corner of block is aligned with BS 267 and WL 139.<br>Style and size of letters are a unit option but will not exceed 3” height. A background block for Pilot/Crew Chief names may be used. The block should be in contrasting color to the section of the aircraft where applied. The WG/CC may approve a locally designed name block to promote unit pride. |
| **NOSE NUMBERS:** | Last four digits, 12” height, located immediately aft of BS 236.30 and centered on WL 187.5. |
| **COMBAT MARKINGS:** | Bomb markings will be enclosed in a 27” by 27”, 1” bordered area centered on BS 218.70 and the bottom edge on WL 139. A conventional bomb or JASSM silhouette may be displayed in the box for each combat sortie flown with a successful weapons release. The JASSM marking will be a maximum of 10” and the conventional bomb silhouette a maximum of 5”. Once the box is full, an existing column of JASSM/Bombs will be removed and replaced with a single JASSM/Bomb silhouette containing a number 10 at the top of the applicable row. |

**Table A2.4. T-38 Aircraft Marking Specifications.**

| T-38 | |
|---|---|
| **COMMAND INSIGNIA:** | 10” centered. Insignia is applied to both sides of the vertical stabilizer.<br>Vertically: Top of insignia 4” below anti-collision light.<br>Horizontally: On line with trailing edge of anti-collision light. |
| **ORGANIZATIONAL INSIGNIA:** | 10” (subdued/CTP, full color AT-38) centered.<br>Vertically: On 3 o'clock position (left) and 9 o'clock position (right).<br>Horizontally: Center of insignia 24” aft of intake lower opening. |
| **DISTINCTIVE UNIT AIRCRAFT IDENTIFICATION MARKING:** | Use two-letter base designation **(Attachment 3)** in 12’’ block letters centered.<br>Vertically: 4” below command insignia.<br>Horizontally: Leading edge of first letter centered on command insignia. |
| **TAIL STRIPE:** | The tail stripe will be applied at the upper portion of the vertical stabilizer and must be in the form of a straight horizontal stripe. The overall height will not exceed 9”. |
| **PILOT / CREW CHIEFS NAMES:** | Pilot: Name centered on left forward canopy rail.<br>Assistant Crew Chief: Name centered on right aft canopy rail. Style and size of letters are a unit option but will not exceed 3” in height. A background block for Pilot/Crew Chief names may be used. The block should be in contrasting color to the section of the aircraft where applied. |
| **NOSE NUMBER:** | 4’’ block numbers. Last three/four digits of the tail number horizontally on both sides of nose gear door. |
| **UNIQUE PAINT SCHEME:** | Companion Trainer Program T-38 aircraft are authorized a gloss gray paint scheme, federal stock code number 16099. |

**Table A2.5. UH-1N Aircraft Marking Specifications.**

| UH-1N | |
|---|---|
| **COMMAND INSIGNIA:** | 10’’ black insignia, placed on right side door, under hinge door window and centered. |
| **ORGANIZATIONAL INSIGNIA (Optional):** | 10’’ black insignia, placed on left side door, under hinge door window and centered. |
| **DISTINCTIVE UNIT AIRCRAFT IDENTIFICATION MARKING:** | Use two-letter base designation **(Attachment 3)** in 10’’black block letters placed 29.5” above the tail stinger cowling edge and 2” from the trailing edge. |
| **PILOT / CREW CHIEFS NAMES (Optional):** | May be applied to either or both pilot and co-pilot’s doors. Style and size of letters are a unit option but will not exceed 4” in height. A background block for Pilot/Crew Chief names may be used. The block may be in contrasting color, using approved UH-1 color scheme to the section of the aircraft where applied. |
| **TAIL STRIPE / ART:** | The tail stripe/art will be applied at the lower portion of the vertical stabilizer, with the bottom of stripe/art no less than 3.5” above stringer cowling. The height of the stripe/art will not exceed 12.5” from the stringer cowling edge. Art will be centered, stripe will span vertical surface. The stripe/art must match gloss requirements of the basic paint scheme, (i.e., aircraft with flat camouflage schemes require application of stripe/art in flat colors). |
| **TAIL NUMBERS:** | The serial numbers will be 5’’ black block numbers, placed 15” above stinger cowling and 2” from trailing edge. |
| **TAIL “USAF” MARKING:** | “USAF” marking will be 5’’ black block letters placed 22.5” above stinger cowling edge with letters grounded 2” from trailing edge. |
| **NOSE NUMBERS (Optional):** | 4’’ numbers. Last four digits of the tail number horizontally centered beneath the glide slope antenna. |
| **SIDE FUSELAGE NUMBERS (Optional):** | 6’’ numbers.<br>Last two digits of the tail number, centered vertically and horizontally between steps on right side only, 2” aft of crew door. |

## Attachment 2 (DYESSAFB)

## STANDARD DYESS AIRCRAFT MARKINGS

**Figure A2.1. (DYESSAFB) Standard Dyess Aircraft Markings (Left Side).**

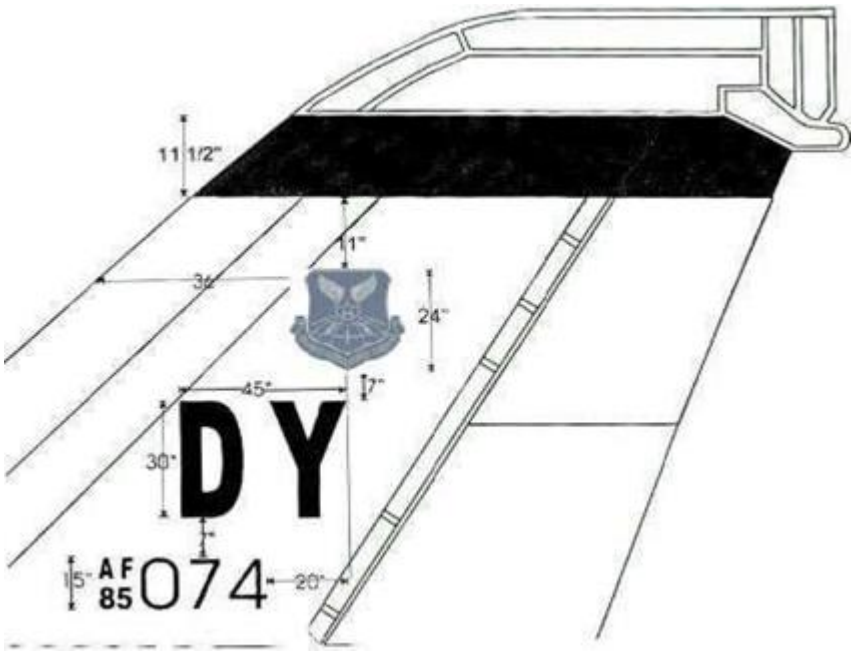



**Figure A2.2. (DYESSAFB) Dyess Command Aircraft Tail Markings (Right Side).**

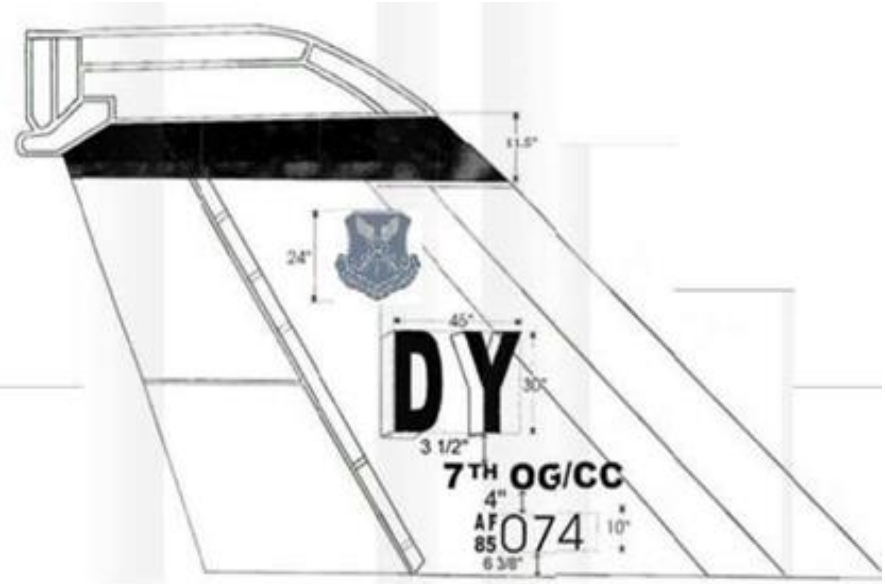


**Figure A2.3. (DYESSAFB) Dyess Command Aircraft Tail Markings with Shadowing (Left Side).**




**Figure A2.4. (DYESSAFB) Dyess Command Aircraft Tail Markings (Right Side).**



**Figure A2.5. (DYESSAFB) Typical 9 BS Tail Markings.**




**Figure A2.6. (DYESSAFB) 28BS Command Tail Marking.**



**Figure A2.7. (DYESSAFB) Wing Insignia below DSO window (Right Side Wing Patch).**

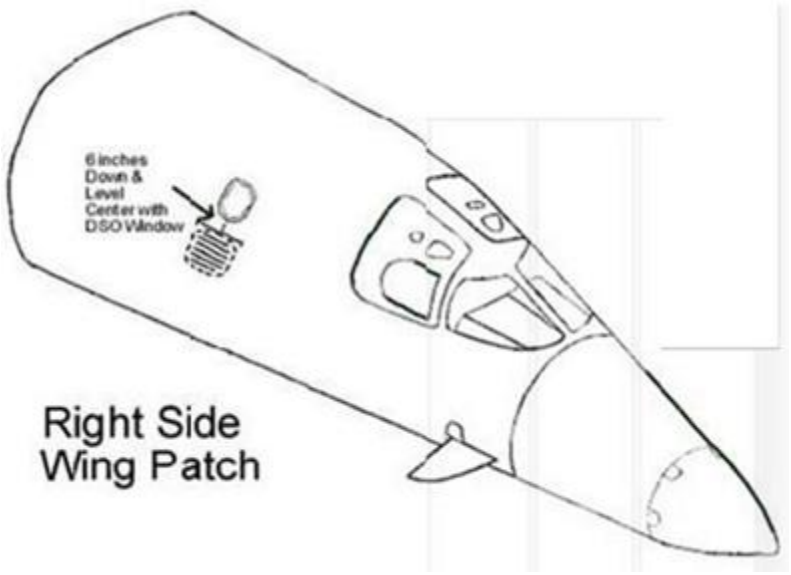



**Figure A2.8. (DYESSAFB) Wing Insignia below WSO Window (In Color).**



**Figure A2.9. (DYESSAFB) Wing Insignia below OSO Window (Left Side Squadron Patch).**

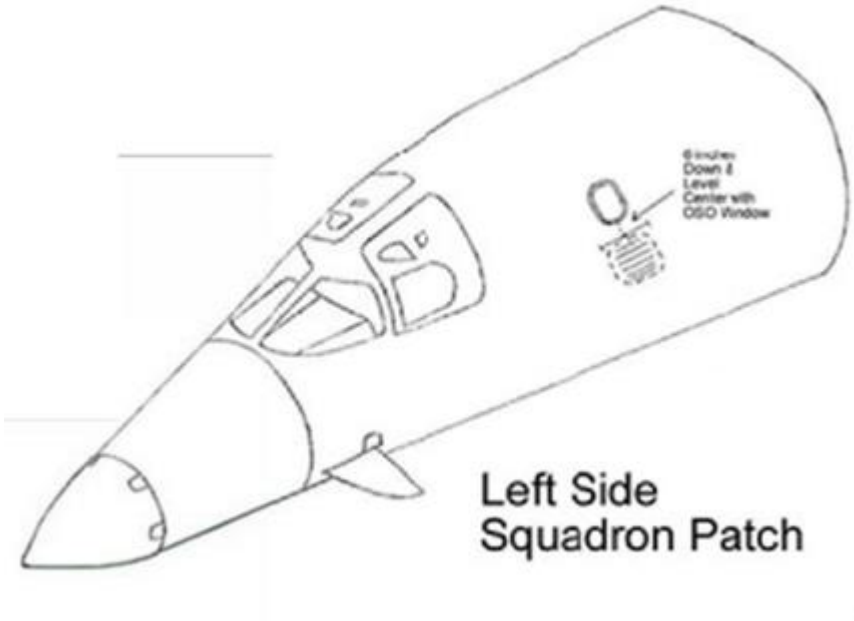



**Figure A2.10. (DYESSAFB) Dyess Command Aircraft Insignia below WSO Window.**




**(NOTE:** Picture above applies to 7 BW/CC only. Other aircraft will have their squadron patch).

**Figure A2.11. (DYESSAFB) Paint Identification Placard.**

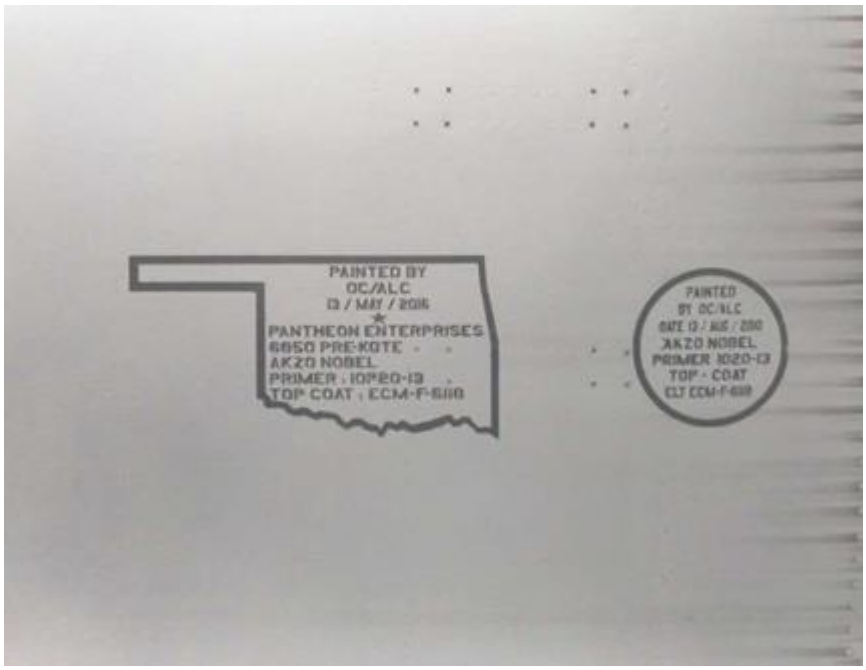



**Figure A2.12. (DYESSAFB) Crew Chief Names.**



**Figure A2.13. (DYESSAFB) 77WPS Tail Marking.**




**Figure A2.14. (DYESSAFB) 77WPS Nose Patch.**



**Figure A2.15. (DYESSAFB) 337TES Standard Tail Marking.**




**Figure A2.16. (DYESSAFB) 337TES Nose Patch.**



**Figure A2.17. (DYESSAFB) 489th Bomb Group Tail Marking.**




**Figure A2.18. (DYESSAFB) 489th Bomb Group Nose Patch.**



**Figure A2.19. (DYESSAFB) 345BS Standard Tail Marking.**




**Figure A2.20. (DYESSAFB) C-130J Tail Marking Configuration 1 of 2.**




All measurements for the blue and the "DYESS" shall be taken from the American Flag. "Dyess" is 36 inches, centered on the US Flag (the Flag is 47 1/4-inches). The blue portion starts 8 inches from the forward edge of the US Flag and extends toward the end of the black lines. The "Dyess" slant is +20 on left; -20 on right. The 10-inch star is centered in the blue field.

**Figure A2.21. (DYESSAFB) C-130 Tail Marking Configuration 2 of 2.**

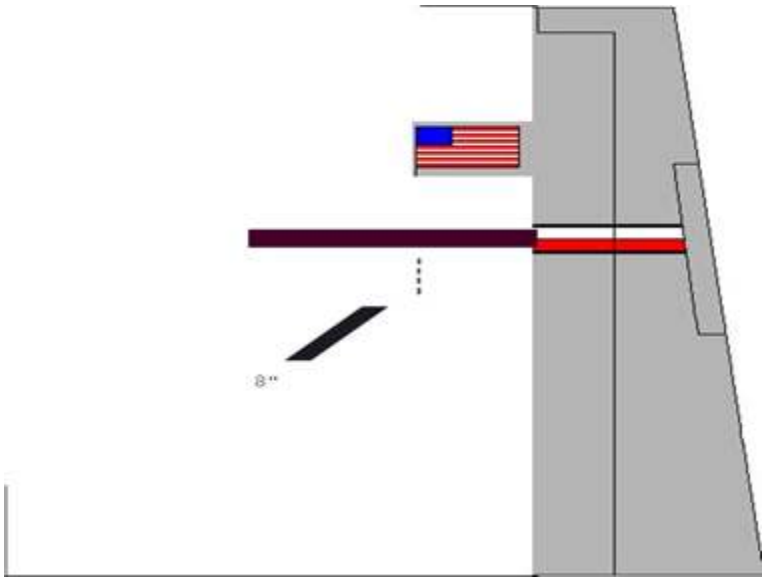
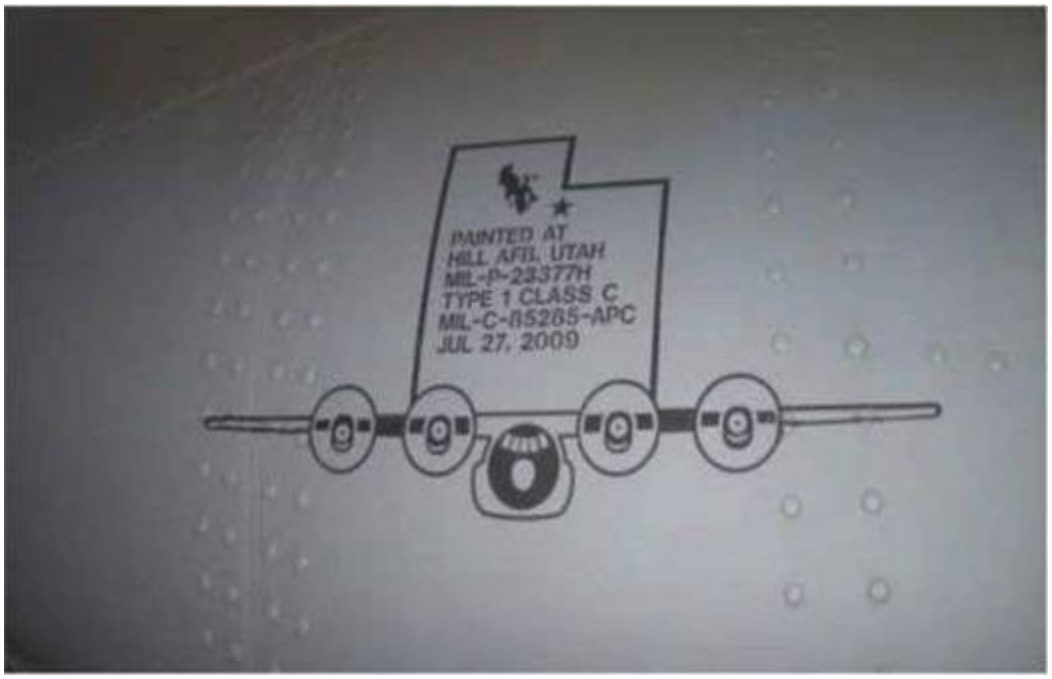



**Figure A2.22. (DYESSAFB) C-130J Paint Date Placard.**



## Attachment 3

## AFGSC DISTINCTIVE UNIT AIRCRAFT IDENTIFICATION MARKINGS LIST

**Table A3.1. AFGSC Distinctive Unit Aircraft Identification Marking List.**

| CODE | AIRCRAFT | UNIT/LOCATION |
|---|---|---|
| DY | B-1B | 7 BW / Dyess AFB, TX |
| EL | B-1B | 28 BW / Ellsworth AFB, SD |
| FE | UH-1N | 582 HG Supporting 90 MW / F.E. Warren AFB, WY |
| LA | B-52H | 2 BW / Barksdale AFB, LA |
| MM | UH-1N | 582 HG Supporting 341 MW / Malmstrom AFB, MT |
| OT | B-52H<br>B-1B | 53 TEG / Barksdale AFB, LA<br>53 TEG / Dyess AFB, TX |
| MT | B-52H<br>UH-1N | 5 BW / Minot AFB, ND<br>582 HG Supporting 91 MW / Minot AFB, ND |
| WA | B-1B | 77 WPS / Dyess AFB, TX |
| WM | B-2A / T-38A | 509 BW / Whiteman AFB, MO |

## Attachment 3 (DYESSAFB)

## B-1B/C-130J AIRCRAFT WASH FACILITY TRAINING/SAFETY BRIEFING

**Figure A3.1. (DYESSAFB) B-1/C-130 Aircraft Wash Facility Training/Safety Briefing.**

**SECTION 1**

1. UPPER TAIL DOOR MUST BE OPEN A MINIMUM OF 12" PRIOR TO OPENING OR CLOSING OF THE LARGE HANGAR DOORS.
2. Fire: Call 117 to report fire. Fire alarms are located on either leading edge of the aircraft wings. If time permits activate the fire alarm furthest away from the fire, and then immediately exit out the closest door. Take a head count, stay safely away, and inform emergency services when they arrive where the fire is located.
3. Personal protective equipment required: face shield or goggles, gloves, rain suit and steel toe boots are mandatory when washing aircraft. They are also mandatory when using Penair HD-1 and HD-4 aircraft cleaning soaps, regardless of what the Safety Data Sheets (SDS) states. Without these protective measures you can expect dermatological issues. This equipment is required when using pressure washers as well as when rinsing the aircraft with water from the fire hoses.
4. SDS are located in the forms on the wall immediately outside of the break room door.
5. Emergency shower/eye wash units are outboard of each wing tip. (**NOTE:** If you wear contact lenses, they must be removed for the duration of the aircraft wash, as soap can cause them to stick to your eyes).
6. The Wash Crew Supervisor is responsible for ensuring a Foreign Object Damage (FOD) walk is accomplished before and after each aircraft wash. Areas to be checked are as follows: hangar bay, including all door tracks with doors open and closed, and aircraft grounding points in the entire facility. The wash crew supervisor will ensure that an applicable hangaring checklist is completed and placed in aircraft forms.
7. Remove all jewelry, rings, necklace, etc., as they can cause harm injury during the wash if caught on an edge or device.
8. Ensure fall protection is used if you are 10 feet or higher from the hangar floor, or as applicable with current guidance.
9. Eating and or drinking is prohibited in the main hangar. Eat only in the break room. Prior to eating, wash your hands to remove possible chemical contamination.
10. A dry pair of shoes and/or clothes are recommended for changing into after the wash. Soap from the wash can cause minor skin irritations if allowed to dwell on your skin.
11. Use only the approved white scrub pads when washing aircraft. Abrasive pads can create deep scratches on aircraft surfaces, creating a path for corrosion to develop.
12. The soap in use is water activated only. Soap is premixed to the proper dilution; increasing soap mixture negatively affects its cleansing ability.
13. The only authorized smoking area is by the Security Forces building across from the wash facility. All doors leading outside have self-closures attached to the door and must not be detached or propped open at any time for any reason.
14. If any outside party wishes to call, the number is (325) 696-2921. Phones are located along the nose side of the hangar and inside the break room.

15. Wash rack parking is for Government Owned Vehicle (GOV) parking only. Ensure a FOD check is accomplished prior to entering the area.
16. Ensure that security signs are placed at the nose and tail of the hangar, immediately following the briefing.
17. Wash cart acquisition is coordinated through AGE at (325) 696-2916 (Ranger 2).
a. A hand receipt is required when equipment is dropped off at the wash rack.
b. The pressure washer is *not* to be used in the wheel well areas.
c. Wash carts can only be used for general washing of aircraft skins and/or cargo bays.
d. Always limit nozzle distance 12-inches to 18-inches from the aircraft surface.
e. These wash units *do not* take the place of hand agitation; they will not remove all of the dirt from the aircraft.
f. The final measure of cleanliness will be reached when guidelines for aircraft washes are met by the applicable -23 TOs.
* **NOTE**: **Power washers can be used on C-130Js only**. Power washers used on B-1Bs **will** cause surface damage and are not authorized by B-1B Structural Engineers (AFLCMC/WWDND).

18. Only trained personnel are authorized to use the pressure washer(s).

a. Please leave the setting AGE has placed on the pressure washer(s), as it is the most effective for its application.
b. Exercise care and caution always when using pressure washers, as injury can occur.

## Section 2

**WASH TEAM SUPERVISOR RESPONSIBILITIES**

1. Ensure entries for wash and lube are entered into the aircraft forms.

2. If the fall arrest system is to be used, ensure that you do an inspection of all devices prior to use.

a. Check the roller for free movement on the I-beams to which they are attached.
b. Check the stitching on all lanyards ensuring the devices are serviceable.
c. Pull test the fall arrest system. To do this, pull firmly on the lanyard, when it locks, ensure it does not pull loose from the I-beam. Move it from side-to-side to ensure the rollers will not disengage.
d. Refer to the user manual for further information (located in the wash rack binder).

**NOTE**: Do not use the fall arrest unit until you initial the AFTO Form 244, *Industrial/ Support Equipment Record*, after accomplishing the prior-to-use inspection.

3. All wash specific TOs, including aircraft specific Series 23 (Material Management), and TO 1- 1- 691, *Aircraft Weapons Systems Cleaning and Corrosion Control,* as well as all applicable work cards are to be provided by the Wash Crew Supervisor.

4. All tools and safety gear (including air hoses) will be provided by the wash crew.

5. All used scrub pads are considered hazardous waste and will be disposed of in hazardous waste Initial Accumulation Points. DO NOT throw them in the trash or leave them at the wash facility.

6. Ensure aircraft has been properly washed, lubed, and signed off in the aircraft forms.

7. All trash and FOD picked up are emptied at the end of each shift and prior to leaving the hangar. This includes the hangar area, the break room, the cigarette can, and the parking area located outside near the hangar nose.

8. Both bathrooms must be cleaned upon completion of wash operations, to include cleaning the sinks and mopping the floors.

9. The break room must be swept and mopped; place all furniture back to its perspective areas after wash operations have been conducted.

10. The Wash Crew Supervisor is responsible for ensuring these tasks are accomplished and will be held accountable for any damage caused by any wash crew member. If any damage has occurred by any Wash Crew Member, it must be reported to the Wash Rack Supervisor or the Corrosion Control shift supervisor immediately.

11. The Wash crew supervisor must contact the 7 EMS Production Superintendent at (325) 518-2546 or the Corrosion Control Shift Supervisor at (325) 696-2168/2169, when wash operations have been completed. A final walk-through and inspection of the wash rack facility and all common areas will occur before the wash crew is released. Ensure that Aircraft Structural Maintenance/Corrosion Control personnel sign the bottom of this sheet to ensure acceptance of the facility.

Wash Crew Supervisor (Printed Name/Signature/Employee #/Date/Phone)

Corrosion Representative (Printed Name/ Signature/Employee #/Date)

## SIGN THE TOP LINE TO THE LEFT WHEN SIGNING THE FACILITY OUT.

**ONCE THE WASH IS COMPLETE AND THE FACILITY IS CLEANED & READY TO BE INSPECTED, CALL 3 2 5 - 696-2168/2169 or 3 2 5 - 696-2240. ASK FOR THE CORROSION SHIFT SUPERVISOR AND LET THEM KNOW THAT YOU ARE READY TO HAVE THE FACILITY SIGNED BACK IN.**

## AFTER DUTY HOURS, CONTACT 7 EMS PRODUCTION SUPERINTENDENT AT 325- 518- 2546.

**ENSURE THAT THE PERSONNEL ACCEPTING RESPONSIBILITY & OWNERSHIP OF THE FACILITY SIGNS THE BOTTOM LINE.**

## Attachment 4

## ICBM PAINT PLAN

**Table A4.1. Paint Plan.**

| Launch Control Center (LCC) | Paint Color Code | Color |
|---|---|---|
| Acoustical enclosure – external | 16473 | Lt Grey |
| Acoustical enclosure – internal | 17925 | White |
| Battery Storage below LCC floors | 16473 | Lt Grey |
| Blast door exterior and racings (except machined surfaces) | 16473 | Lt Grey |
| Blast door interior | 16473 | Lt Grey |
| Cable trays | 16473 | Lt Grey |
| Ducts, pipes and conduits | Same as associated equipment/fixtures | |
| Electrical Surge Arrestor (ESA) room walls/floor | 16473 | Lt Grey |
| Equipment racks | 24525 | Lt Green |
| ESA room panels, door and trim | 16473 | Lt Grey |
| External metal racks | 24525 | Lt Green |
| LCC liner | 16473 | Lt Grey |
| Miscellaneous tanks and brackets | Same as associated equipment/fixtures | |
| Shock isolators | 17925 | White |
| Tunnel junction floor | 16473 | Lt Grey |
| Tunnel junction walls | 17925 | White |
| Unit Reference Designator (URD) background | 23538 | Yellow |
| URD stencils | 17038 | Black |
| **Launch Control Equipment Building** | | |
| Air conditioning equipment | 16473 | Lt Grey |
| Automatic switching unit | 16473 | Lt Grey |
| Diesel engines and switch gear | Per Manufacturer | Per Manufacturer |
| Diesel start battery rack | 17038 | Black |
| Electrical panels - exterior | 16473 | Lt Grey |
| Floors | 16473 | Lt Grey |
| Tanks, Ducts, pipes and conduits | Same as associated equipment/fixtures | |
| URD background | 23538 | Yellow |
| URD stencils | 17038 | Black |
| Walls | 17925 | White |
| **Launch Control Support Building** | | |
| Air conditioning equipment | 16473 | Lt Grey |
| Automatic switching unit | 16473 | Lt Grey |
| Diesel engines and switch gear | Per Manufacturer | Per Manufacturer |

| Launch Control Center (LCC) | Paint Color Code | Color |
|---|---|---|
| Diesel start battery rack | 17038 | Black |
| Electrical panels - exterior | 16473 | Lt Grey |
| Floors | 16473 | Lt Grey |
| Tanks, Ducts, pipes and conduits | Same as associated equipment/fixtures | |
| URD background | 23538 | Yellow |
| URD stencils | 17038 | Black |
| Walls | 17925 | White |
| **Launch Facility Support Building** | | |
| Air conditioning equipment | 16473 | Lt Grey |
| Diesel engine and switch gear | Per Manufacturer | |
| Diesel start battery racks | 17038 | Black |
| Electrical panels | 16473 | Lt Grey |
| Exhaust fan | 16473 | Lt Grey |
| Exposed metal and hangar supports | Same as associated equipment/fixtures | |
| Floors | | Red Oxide Primer |
| Instrument air compressor | 16473 | Lt Grey |
| Pipes, conduits and tanks | Same as associated equipment/fixtures | |
| Repeater telephone set | 24525 | Lt Green |
| Under floor ducting | 16473 | Lt Grey |
| URD background | 23538 | Yellow |
| URD stencils | 17038 | Black |
| Walls | 17925 | White |
| **Launch Facility Topside** | | |
| Air intake vent plate | 16473 | Lt Grey |
| Clutter plate mast | 16473 | Lt Grey |
| Collimator Bench | 17925 | White |
| Diesel fill cover (metal or fiberglass) | 16473 | Lt Grey |
| Diesel fill pipe and cap | 16473 | Lt Grey |
| Hatch covers, grates and ladders | 16473 | Lt Grey |
| IMPSS antenna | 17925 | White |
| Launcher closure arrestors (Vandenberg) | 16473 | Lt Grey |
| Launcher closure door (top trim) | 16473 | Lt Grey |
| Launcher closure door (front, metal portion) | 16473 | Lt Grey |
| Launcher closure door revetment walls (Vandenberg) | 16473 | Lt Grey |
| Launcher closure thermal shield | 36440 | Non-skid Grey |
| Launcher closure thermal shield alternate coating | 16473 | Non-skid Grey |
| Launch facility support building door/frame | 16473 | Lt Grey |
| Miscellaneous exposed metal surfaces | 16473 | Lt Grey |

| Launch Control Center (LCC) | Paint Color Code | Color |
|---|---|---|
| Primary access hatch hand driven linear actuator | 16473 | Lt Grey |
| Security pit and cover | 16473 | Lt Grey |
| A-Vault door | 15045 | Blue |
| Sump pump drain line | 16473 | Lt Grey |
| Transporter-erector pylons | 16473 | Lt Grey |
| Transporter-erector tie-down slots | 16473 | Lt Grey |
| UHF radio marker posts | 17925 | White |
| URD background | 23538 | Yellow |
| URD stencils | 17038 | Black |
| **Launch Facility Equipment Rooms** | | |
| Walls and ceiling | 17925 | White |
| Ballistic actuator Piston Rod Support | 15045 | Blue |
| Launch tube heater | 16473 | Lt Grey |
| Umbilical retract | 15045 | Blue |
| Sump pump (SP-102) control box | No coating authorized | |
| Work cage distribution box | 15045 | Blue |
| Electro-Mechanical Actuator | 15045 | Blue |
| Motor generator | 15045 | Blue |
| Battery support brackets | 17925 | White |
| Launcher closure ballistic actuator | 15045 | Blue |
| Air handler control panel | 16473 | Lt Grey |
| Emergency fan | 15045 | Blue |
| Safety stripes | 23538 | Yellow |
| Air handler drain line protection ramp | 23538 | Yellow |
| Launch tube heater control panel | 16473 | Lt Grey |
| Emergency fan air duct | 15045 | Blue |
| Secondary door | 24300 | Sea Foam Green |
| Ladder (1st to 2nd level) | 17925 | White |
| Monorail I-beam assembly | 17925 | White |
| Shock isolators | 24300 | Sea Foam Green |
| Shock isolated floor movement limit lines | 17038 | Black |
| Primary door (interior) | 17925 | White |
| Primary door (exterior band) | 16473 | Lt Grey |
| Personnel access shaft | 17925 | White |
| Primary door bearing surface | 17925 | White |
| Pipes and conduits | Same as associated equipment/fixtures | |
| Launcher distribution panel | 16473 | Lt Grey |
| Communications jack boxes | 17925 | White |
| Rattle space EMI screen | IAW TO 21M-LGM30F-101 | |
| Rattle space floor under foam blocks | 16473 | Lt Grey |
| Floors | 16473 | Lt Grey |
| ESA room walls/floor | 16473 | Lt Grey |

| Launch Control Center (LCC) | Paint Color Code | Color |
|---|---|---|
| ESA room panels, door and trim | 16473 | Lt Grey |
| Air conditioner equipment (New) | 16473 | Lt Grey |
| Electro-Mechanical Actuator support racks | 16473 | Lt Grey |
| Equipment rack air duct | 17925 | White |
| J-Ladder platform | No coating authorized | |
| LER Safety Posts | 23538 | Yellow |
| URD background | 23538 | Yellow |
| URD stencils | 17038 | Black |
| **Launch Tube** | | |
| Launch tube steel liner | 17925 | White |
| Launch tube floor | 17925 | White |
| Pipes and conduits | 17925 | White |
| Missile suspension system travel limit markings | 17038 | Black |
| Missile suspension system and components | 15045 | Blue |

## Attachment 4 (DYESSAFB)

## PAINT SCORING FOR B-1B/C-130J AIRCRAFT WEAPONS SYSTEMS

**Figure A4.1. (Added-DYESSAFB) Paint Scoring For B-1B/C-130 Aircraft Weapons Systems.**

| Aircraft Tail Number A0365 | Intakes | Wing Slats | Fuselage | Stabilator, Leading Edge Tape | Total Score Value in Highest Need | Remarks |
|---|---|---|---|---|---|---|
| | 1 | 5 | 4 | 3 | 13 | |

| |
|---|
| **0** for individual component score = No defects (**Nothing required**) |
| **1** for individual component score = Negligible (**Nothing required**) Total score of 0 through 4 is Negligible -- **Nothing required** |
| **2** for individual component score = Minor paint (*Brush and or spray touch-up can be used*) Total score of 5 through 8 is Minor Paint -- **Brush and or spray touch-up can be used** |
| **3** for individual component score = Moderate paint (*Spray paint required in area*) Total score of 9 through 12 is Moderate Paint -- **Spray paint required areas** |
| **4** for individual component score = Extensive paint (*Spray paint required in area*) Total score of 13 through 16 is Extensive Paint -- **Spray paint required areas** |
| **5** for individual component score = Corrosion problem (*Major paint required*) Total score of 17 through 20+ is Corrosion problem -- **Major paint required** |

**Figure A4.2. (Added-DYESSAFB) B-1B Paint Score Sheet.**

**B-1B Paint Score Sheet**

**AIRCRAFT TAIL NUMBER**_____

**NOSE ART**_____Applied Yes or No (**Verify Nose Art Approval Letter located at:** *7mxg_ems_ws/FAB/Wing Corrosion Manager/B-1B Nose Art)*

**PAINT DATE**_____**SCORED BY**_____

**DATE SCORED**_____**TOTAL COMPOSITE SCORE**_____

**JCN**_____

| |
|---|
| **0** for individual component score = No defects (*Nothing required)* |
| **1** for individual component score = Negligible (*Nothing required)* |
| Total composite score of 0 through 4 is Negligible -- **Nothing required** |
| **2** for individual component score = Minor paint (*Brush and or spray touch-up can be used*) |
| Total composite score of 5 through 8 is Minor paint -- **Brush and or spray touch-up can be used** |
| **3** for individual component score = Moderate paint (*Spray paint required in area*) |
| Total composite score of 9 through 12 is Moderate paint -- **Spray paint required in areas** |
| **4** for individual component score = Extensive paint (*Spray paint required in area*) |
| Total composite score of 13 through 16 is Extensive paint -- **Spray paint required in areas** |
| **5** for individual component score = Corrosion problem (*Major paint required)* |
| Total composite score 17 through 20 is Corrosion problem -- **Major paint required** |

**WINGS** _____

**(CIRCLE ONE)** **0** **1** **2** **3** **4** **5**

**FUSELAGE** _____

**(CIRCLE ONE)** **0** **1** **2** **3** **4** **5**

**ENGINES**_____

**(CIRCLE ONE)** **0** **1** **2** **3** **4** **5**

**EMPENNAGE** _____

**(CIRCLE ONE)** **0** **1** **2** **3** **4** **5**

**(ANNOTE ANY DISCREPENCIES ON DIAGRAMS)**

Aircraft Tail #_____ Date Scored_____ Scored By_____

**Figure A4.3. (Added-DYESSAFB) B-1B.**

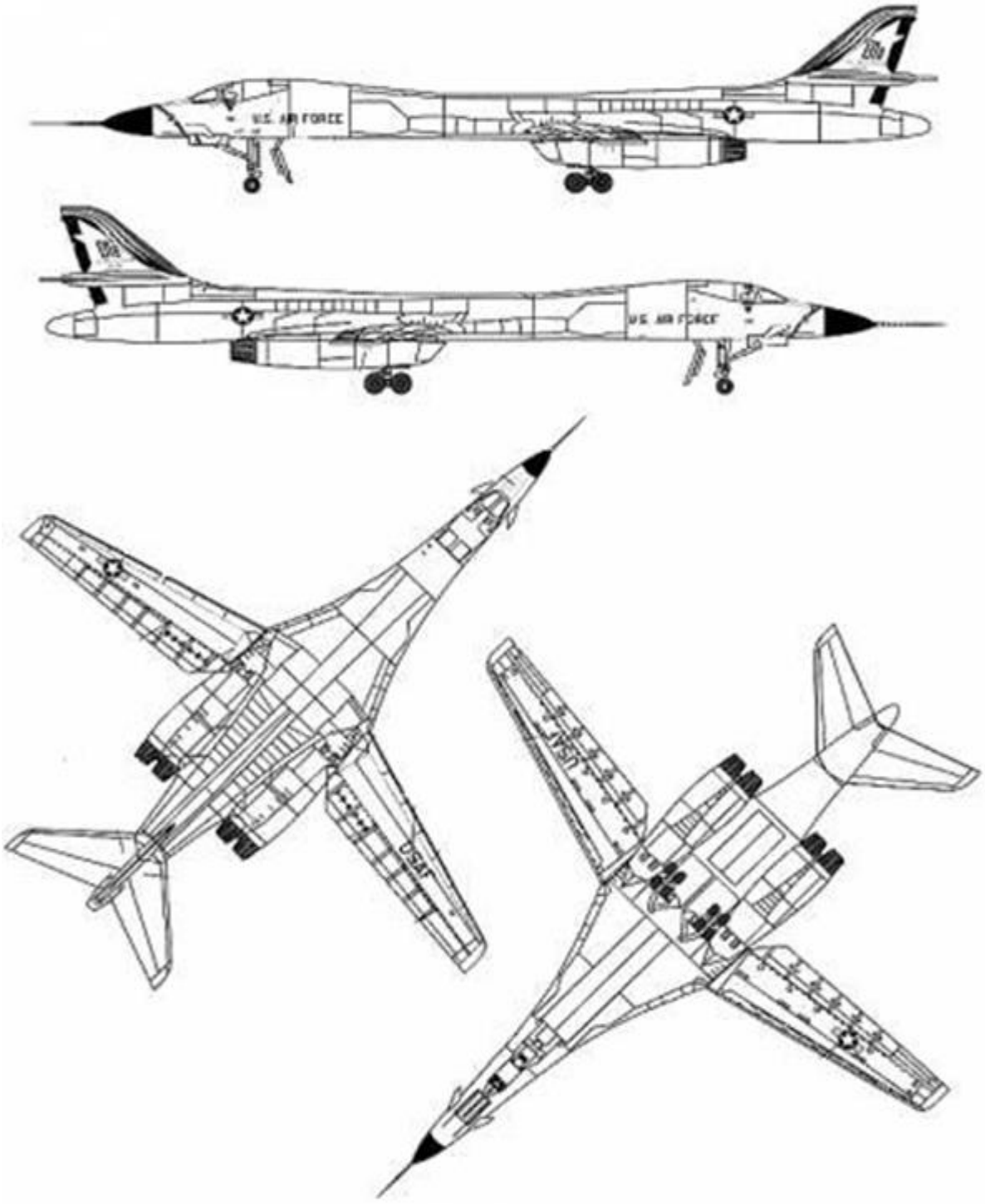


**Figure A4.4. (Added-DYESSAFB) C-130J Paint Score Sheet.**

| |
|---|
| **C-130J Paint Score Sheet**<br><br>**AIRCRAFT TAIL NUMBER_____**<br><br>**NOSE ART__**Applied Yes or No<br><br>**PAINT DATE_____**<br><br>**SCORED BY_____**<br><br>**DATE SCORED____**<br><br>**TOTAL COMPOSITE SCORE_____**<br><br>**JCN_____** |
| **0** for individual component score = No defects (*Nothing required)* |
| **1** for individual component score = Negligible (*Nothing required)*<br>Total composite score of 0 through 4 is Negligible **-- Nothing required** |
| **2** for individual component score = Minor paint (*Brush and or spray touch-up can be used*)<br>Total composite score of 5 through 8 is Minor paint **-- Brush and or spray touch-up can be used** |
| **3** for individual component score = Moderate paint (*Spray paint required in area*)<br>Total composite score of 9 through 12 is Moderate paint **-- Spray paint required in areas** |
| **4** for individual component score = Extensive paint (*Spray paint required in area*)<br>Total composite score of 13 through 16 is Extensive paint **-- Spray paint required in areas** |
| |

| |
|---|
| **5** for individual component score = Corrosion problem (*Major paint required)* |
| Total composite score 17 through 20 is Corrosion problem **-- Major paint required.**<br>**WINGS_____** |
| **(CIRCLE ONE) 0 1 2 3 4 5**<br>**FUSELAGE _____** |
| **(CIRCLE ONE) 0 1 2 3 4 5**<br>**ENGINES_____** |
| **(CIRCLE ONE) 0 1 2 3 4 5**<br>**(ANNOTE ANY DISCREPENCIES ON DIAGRAMS)**<br>**Aircraft Tail #_____Date Scored_____**<br>**Scored By _____** |

**Figure A4.5. (Added-DYESSAFB) C-130 (1 of 2).**

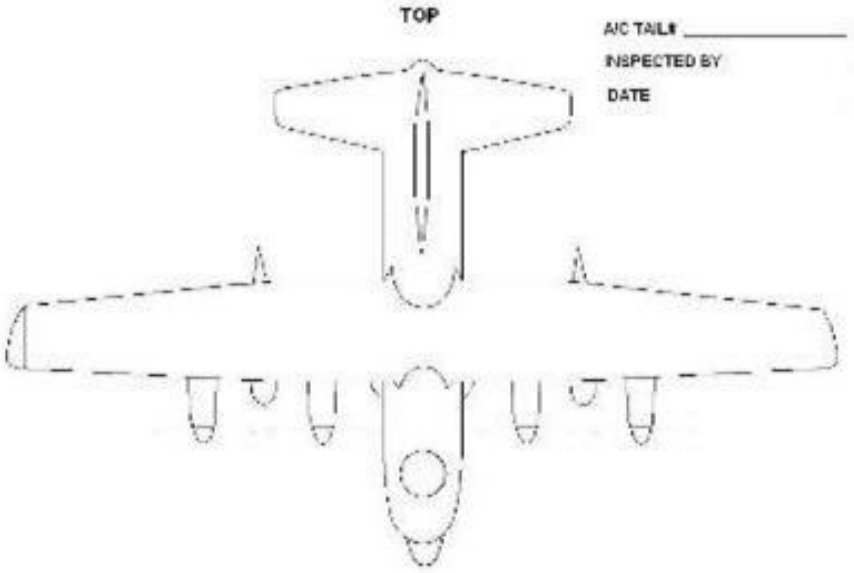



**Figure A4.6. (Added-DYESSAFB) C-130J (2 of 2).**

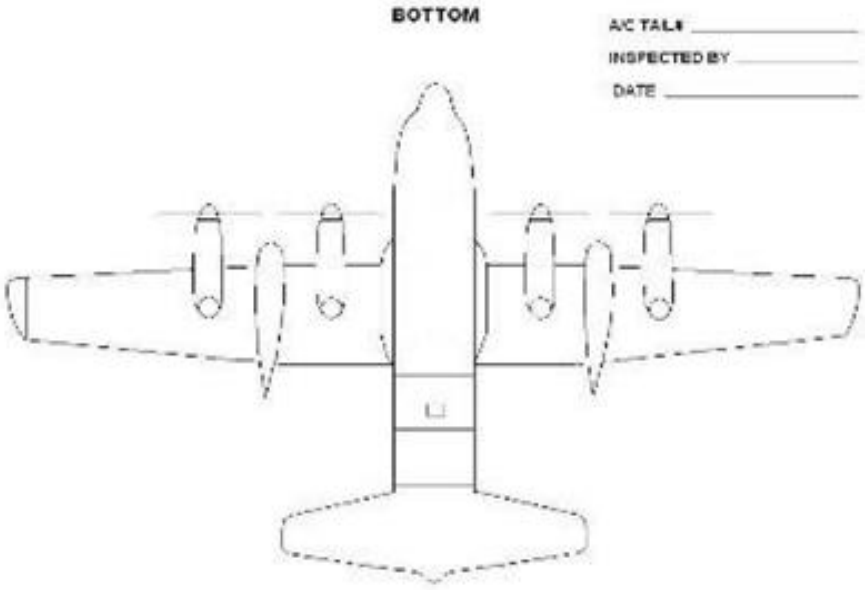


**Figure A4.7. (Added-DYESSAFB) C-130 Left and Right Side.**

